

$$\frac{d}{dx} \int_0^x f(t) dt = f(x), > 0$$

$$\frac{d}{dx} F(x) = \frac{\cancel{f(x)} \int_0^x f(t) dt - \cancel{f(x)} \int_0^x t f(t) dt}{\left(\int_0^x f(t) dt \right)^2}$$

$$f(x) > 0$$

$$\Rightarrow \int_0^x f(t) dt > 0.$$

$$= \frac{f(x) \int_0^x (x-t) f(t) dt}{\left(\int_0^x f(t) dt \right)^2} > 0.$$

例 5.3.5. 设 $f(x)$ 在 $[a, b]$ 上连续且 $f'(x) > 0$, $f(x)$ 在 (a, b) 可导,

$$\int_a^b f(x) dx < \frac{f(a) + f(b)}{2} (b-a)$$

证法:

$$x_1, x_2 \in [a, b] \quad f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1) + f(x_2)}{2}$$

$f'(x) > 0$.

$$\text{令 } g(x) = \left[2 \int_a^x f(t) dt \right] - [f(a) + f(x)] (x-a) \quad x \in [a, b].$$

当 $x=a$

$$g(a) = 0.$$

$$g(b) < 0.$$

$$\forall x \in (a, b)$$

$$g'(x) = 2f(x) - (f(a) + f(b)) - f'(x)(x-a)$$

$$= f(x) - f(a) - f'(x)(x-a)$$

$$= f'(\xi)(x-a) - f'(x)(x-a) \quad \xi \in (a, x)$$

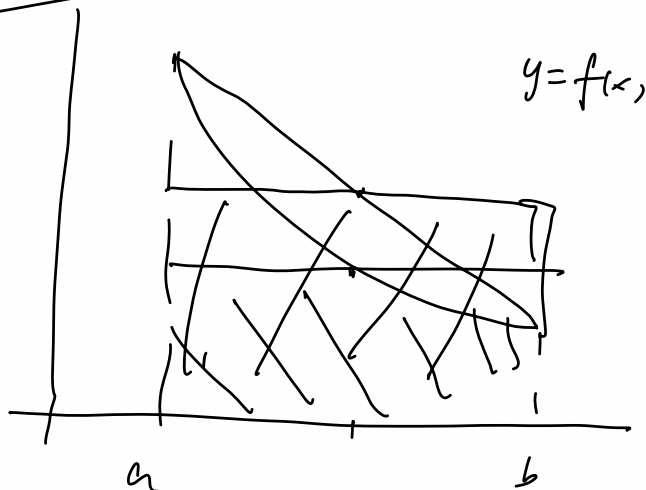
$$= (f'(\xi) - f'(x))(x-a) \quad \begin{matrix} < 0 & > 0 \end{matrix}$$

$$\Rightarrow g(x) \text{ is decreasing}$$

$$\Rightarrow g(b) < g(a) \Rightarrow 0$$

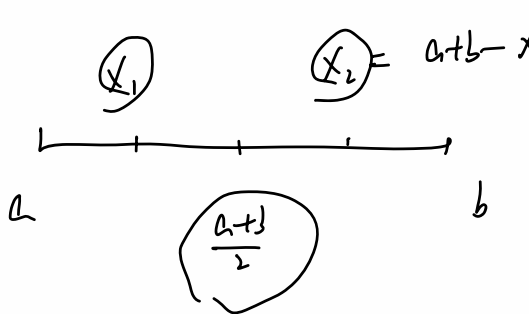
$$\underline{f\left(\frac{a+b}{2}\right)}$$

$$< \frac{1}{b-a} \int_a^b f(x) dx < \frac{f(a) + f(b)}{2}$$



$$\int_a^b f(x) dx > (b-a) f\left(\frac{a+b}{2}\right)$$

$$\forall x_1, x_2 \in [a, b]. \quad f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$$



$$\frac{x_1+x_2}{2} = \frac{a+b}{2}$$

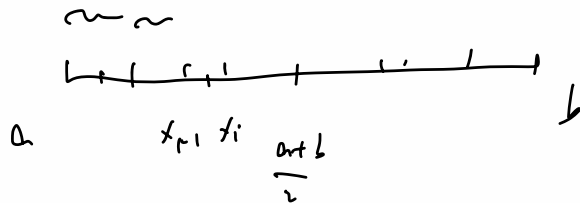
$$(b-a) f\left(\frac{a+b}{2}\right) < \frac{f(x_1)+f(x_2)}{2} = \frac{f(x_1)^{\frac{b-a}{2}} + f(x_2)^{\frac{b-a}{2}}}{2}$$

2n 等分.

$$x_0 = a$$

$$x_i = a + \frac{i}{2n} (b-a), \quad i=1, 2, \dots, 2n,$$

$$\xi_i = \frac{x_{i-1}+x_i}{2} \in [x_{i-1}, x_i]$$



$$\begin{aligned} \xi_1 + \xi_{2n} &= \frac{x_0+x_1}{2} + \frac{x_{2n-1}+x_{2n}}{2} \\ &= a+b. \end{aligned}$$

$$\xi_i + \xi_{2n+1-i} = a+b$$

$$\sum_{i=1}^{2n} f(\xi_i) \frac{b-a}{2n} = \frac{b-a}{n} \sum_{i=1}^{2n} f(\xi_i) \frac{1}{2}$$

$$= \frac{b-a}{n} \left[\frac{f(\xi_1)+f(\xi_{2n})}{2} + \frac{f(\xi_2)+f(\xi_{2n-1})}{2} + \dots \right]$$

$$\begin{aligned}
 &> \frac{b-a}{2} f\left(\frac{a+b}{2}\right) \\
 &= (b-a) f\left(\frac{a+b}{2}\right)
 \end{aligned}$$

例: 设 $x_0 = 25$, $f_n = \arctan x_{n-1}$ ($n=1, 2, \dots$)
证明 $\{x_n\}$ 收敛, 并求极限.

证明: $x_n = f(x_{n-1})$ $f(x) = \arctan x$

$$x_1 < \frac{\pi}{2} < x_0$$

$$x_n > 0.$$

$\{x_n\}$ 是 $\downarrow$.

$$\boxed{\arctan x < x} \quad \frac{1}{2}x > \arctan x > \frac{1}{3}x$$

$$\frac{1}{1+x^2} < 1.$$

$$\int_0^x \frac{1}{1+t^2} dt < \int_0^x 1 dt$$

$$\boxed{\arctan x < x.}$$

$$\lim_{n \rightarrow \infty} x_n = a$$

例: 证明当 $x > 0$ 时

$$x - \frac{x^3}{6} < \sin x < x - \frac{x^3}{6} + \frac{x^5}{120}$$

证: 用

$$\cos x \leq 1 \quad x > 0, \quad (\cos x = 1 \Leftrightarrow x = 2k\pi)$$

$$\int_0^x \cos t \, dt < \int_0^x 1 \, dt$$

$$\Leftrightarrow \sin x < x \quad (x > 0)$$

$$\int_0^x \sin t \, dt < \int_0^x t \, dt$$

$$\Leftrightarrow 1 - \cos x < \frac{1}{2}x^2 \quad (x > 0)$$

$$\int_0^x (1 - \cos t) \, dt < \int_0^x \frac{1}{2} t^2 \, dt$$

$$\Leftrightarrow x - \sin x < \frac{1}{6} x^3$$

$$\Leftrightarrow x - \frac{1}{6} x^3 < \sin x$$

§ 合部积分.

$u(x)$ 和 $v(x)$ 在 $[a, b]$ 上有连续的导数.

$$u dv = d(\underline{uv}) - v du.$$

$$\int_a^b u(x) v'(x) dx = \underline{\underline{\left[u(x) v(x) \right]_a^b}} - \int_a^b u'(x) v(x) dx.$$

$$\text{13.1: } \int_0^{\pi} x \cos x dx$$

$$= \int_0^{\pi} x d \sin x = \left(x \sin x \right) \Big|_0^{\pi} - \int_0^{\pi} \sin x dx$$

$$= \cos x \Big|_0^{\pi} = -2.$$

$$\text{12.1: } \int_0^{\frac{1}{2}} \overset{u}{\arcsinh x} \overset{v.}{dx}$$

$$= x \arcsinh x \Big|_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x d \arcsinh x$$

$$= \frac{\pi}{12} - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx = \frac{\pi}{12} + \frac{1}{2} \int_0^{\frac{1}{2}} \frac{(-2x)}{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{12} + \int_0^{\frac{1}{2}} \frac{1}{2\sqrt{1-x^2}} d(1-x^2),$$

$$= \frac{\pi}{12} + \left. \sqrt{1-x^2} \right|_0^{\frac{1}{2}}$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$$

例: $\int_0^{\frac{\pi}{4}} \frac{x}{1+\cos^2 x} dx$

$$= \int_0^{\frac{\pi}{4}} \frac{x}{2\cos^2 x} dx = \int_0^{\frac{\pi}{4}} \frac{x}{2} d \tan x$$

$$= \left. \frac{x \tan x}{2} \right|_0^{\frac{\pi}{4}} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \tan x dx$$

$$= \frac{\pi}{8} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{\sec x \tan x}{\sec x} dx$$

$(\sec x)'$
 $= \sec x \tan x$

$$= \frac{\pi}{8} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{1}{\sec x} d \sec x$$

$$= \frac{\pi}{8} - \frac{1}{2} \left. \ln \sec x \right|_0^{\frac{\pi}{4}} = \frac{\pi}{8} - \frac{\ln 2}{4}.$$

例 5.4.2. 设 $f(x) = \int_0^x \frac{\sin t}{\pi - t} dx$, 求 $\int_0^\pi f(x) dx$.

解:

$$f'(x) = \frac{\sin x}{\pi - x}.$$

$$f(0) = \int_0^0 \dots = 0.$$

$$f(\pi) = \int_0^\pi \frac{\sin t}{\pi - t} dx > 0.$$

$$\int_0^\pi f(x) dx = \int_0^\pi f(x) d(x - \pi)$$

$$= \underline{\underline{f(x)(x-\pi) \Big|_0^\pi}} - \int_0^\pi (x-\pi) df(x)$$

$t \in (0, \pi), \quad \sin t > 0,$
 $\pi - t > 0.$

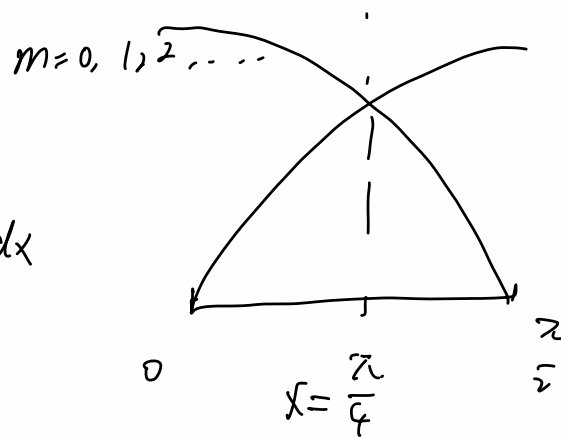
$$d(x - \pi) = dx$$

$$= - \int_0^\pi (x - \pi) f'(x) dx$$

$$= \ominus \int_0^\pi \underbrace{(x - \pi)}_{< 0} \underbrace{\frac{\sin x}{\pi - x}}_{> 0} dx = \int_0^\pi \sin x dx = 2.$$

例 5.4.3. 求证 $\int_0^{\frac{\pi}{2}} \cos^m x dx = \int_0^{\frac{\pi}{2}} \sin^m x dx, \quad m=0, 1, 2, \dots$

$$\underline{\underline{I_m = \int_0^{\frac{\pi}{2}} \cos^m x dx = \int_0^{\frac{\pi}{2}} \sin^m x dx}}$$



$$I_0 = \frac{\pi}{2} \quad I_1 = 1.$$

$$\text{if } m \geq 2. \quad I_m = \int_0^{\frac{\pi}{2}} \cos^m x \, dx = \underbrace{\int_0^{\frac{\pi}{2}} \cos^{m-1} x \, d \sin x.}$$

$$= \cos^{m-1} x \sin x \Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, d(\cos^{m-1} x)$$

$$= + \int_0^{\frac{\pi}{2}} \sin x \cdot (m-1) \cos^{m-2} x \cdot (-\sin x) \, dx$$

$$= (m-1) \int_0^{\frac{\pi}{2}} \sin^2 x \cos^{m-2} x \, dx$$

$$= (m-1) \int_0^{\frac{\pi}{2}} \left(\underbrace{\cos^{m-2} x} - \underbrace{\cos^m x} \right) dx$$

$$I_m = (m-1) I_{m-2} - (m-1) I_m$$

$$I_m = \frac{m-1}{m} I_{m-2}, \quad m \geq 2.$$

$$\frac{I_m = () () () 1}{I_m = () () () \frac{\pi}{2}}$$

$$k \geq 1.$$

$$\left\{ \begin{aligned} (2k-1)!! &= (2k-1)(2k-3) \cdots 3 \cdot 1 \\ (2k)!! &= 2k(2k-2)(2k-4) \cdots 4 \cdot 2 \end{aligned} \right.$$

Wallis
公式.

$$\begin{cases} I_{2k-1} = \frac{(2k-2)!!}{(2k-1)!!} \\ I_{2k} = \frac{(2k-1)!!}{(2k)!!} \cdot \frac{\pi}{2} \end{cases} \quad k=1,2,\dots$$

$$\lim_{n \rightarrow \infty} \int_0^{\frac{\pi}{2}} \sin^n x \, dx \geq 0.$$

Wallis 公式: $\lim_{n \rightarrow \infty} \frac{(n!)^2 2^{2n}}{(2n!)^2 \sqrt{n}} = \sqrt{\pi}$

$\frac{1}{2} \pi \in (0, \frac{\pi}{2})$, $0 < \sin x < 1$.

$$\sin^{2n+1} x < \sin^{2n} x < \sin^{2n-1} x$$

$$I_{2n+1} < I_{2n} < I_{2n-1}$$

$$\frac{(2n)!!}{(2n+1)!!} < \frac{(2n-1)!!}{(2n)!!} \cdot \frac{\pi}{2} < \frac{(2n-2)!!}{(2n-1)!!}$$

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n+1)!!} \right)^2 \frac{\pi}{2} < \frac{\pi}{2}$$

Stirling 公式:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad (n \rightarrow \infty)$$

例 5.44. (积分余项)

设 $f(x)$ 在 $[a, b]$ 有 $n+1$ 阶连续导数, $\forall x_0 \in [a, b]$.

$$f(x) = P_n(x; x_0; n) + R_n(x)$$

$$R_n(x) = \frac{1}{n!} \int_{x_0}^x (x-t)^n f^{(n+1)}(t) dt$$

$f'(x)$

证明:

$$\int_{x_0}^x f'(t) dt = f(x) - f(x_0)$$

$$\int_{x_0}^x + \int_a^x + \int_{x_0}^a$$

$$f(x) = f(x_0) + \int_{x_0}^x f'(t) dt.$$

$$= f(x_0) + \int_{x_0}^x f'(t) d(t-x_0)$$

$$= f(x_0) + f'(x_0)(x-x_0) - \int_{x_0}^x \frac{(t-x_0)^2}{2} f''(t) dt$$

$$= f(x_0) + f'(x_0)(x-x_0) - \left[\frac{(t-x_0)^2}{2} f''(t) \right]_{x_0}^x$$

$$\frac{1}{2} (t-x_0)^2$$

$$- \frac{1}{2} \int_{x_0}^x \underline{(t-x)^2} f''(t) dt \quad \overline{3}(t-x)$$

$$= f(x_0) + f'(x_0)(x-x_0) + \frac{(x-x_0)^2}{2} f''(x_0) \\ + \frac{1}{3!} (t-x)^3 f'''(t) \Big|_{x_0}^x - \frac{1}{3!} \int_{x_0}^x (t-x)^3 f^{(4)}(t) dt.$$

⋮

$$= \dots + \frac{1}{n!} \int_{x_0}^x (x-t)^n f^{(n+1)}(t) dt.$$

$$\frac{1}{n+1} f^{(n+1)}(\xi) (x-x_0)^{n+1}$$

$$t \in [x_0, x].$$

$$x-t \geq 0.$$

$$\int_{x_0}^x \underbrace{(x-t)^n}_f \underbrace{f^{(n+1)}(t)}_g dt$$

$$= f^{(n+1)}(\xi) \int_{x_0}^x (x-t)^n dt \quad \xi \in (x_0, x)$$

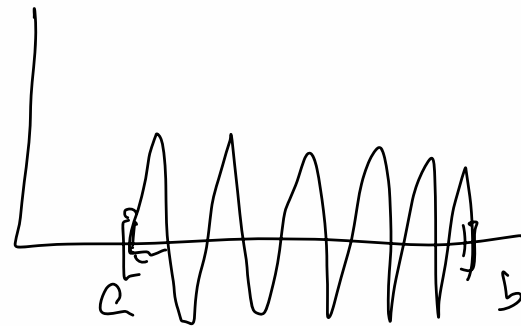
$$= f^{(n+1)}(\xi) \left[-\frac{1}{n+1} (x-x_0)^{n+1} \right]_{x_0}^x$$

$$= f^{(n+1)}(\xi) \cdot \frac{1}{n+1} (x-x_0)^{n+1}$$

证: Riemann - Lebesgue 定理.

设 $f(x)$ 在 $[a, b]$ 有一阶连续导数.

$$\lim_{p \rightarrow +\infty} \int_a^b f(x) \sin px \, dx = 0.$$



$$\begin{aligned} \text{证: } \int_a^b f(x) \sin px \, dx &= \int_a^b f(x) \, d\left(-\frac{1}{p} \cos px\right) \\ &= \underbrace{f(x) \left(-\frac{1}{p} \cos px\right) \Big|_a^b} + \underbrace{\int_a^b \left(\frac{1}{p} \cos px\right) f'(x) \, dx}_{\leq \frac{M}{p}} \\ &= \frac{1}{p} \left(\underbrace{f(a) \cos pa - f(b) \cos pb}_{\leq \frac{M}{p}} \right) \\ &\quad + \frac{1}{p} \underbrace{\left(\int_a^b f'(x) \cos px \, dx \right)}_{\leq \frac{M}{p}} \end{aligned}$$

$$|f(a) \cos pa - f(b) \cos pb| \leq |f(a)| \underbrace{|\cos pa|} + |f(b)| \underbrace{|\cos pb|}$$

$$\leq \underbrace{|f(a)| + |f(b)|}_{\leq 2}$$

$$\left| \int_a^b f'(x) \cos px \, dx \right| \leq \int_a^b |f'(x) \cos px| \, dx$$

M

$$\leq \int_a^b \underbrace{|f'(x)|}_{\text{可积}} \, dx$$

$f(x)$ $\hookrightarrow$
 $f'(x)$ 可积

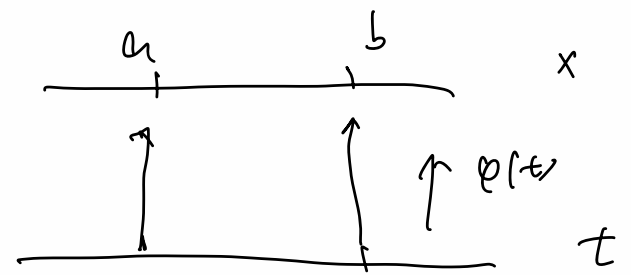
§ 换元法.

定义: $f(x)$ 在 I 区间上连续, 且 $a, b \in I$, $\varphi(t)$ 在 $[\alpha, \beta]$ 或 $[\beta, \alpha]$ 上有连续的导数

$$\begin{cases} \varphi([\alpha, \beta]) \subseteq I \\ \varphi([\beta, \alpha]) \subseteq I \end{cases}$$

且 $\varphi(\alpha) = a$, $\varphi(\beta) = b$ 则:

$$\int_a^b f(x) \, dx = \int_{\alpha}^{\beta} \underline{\underline{f(\varphi(t)) \varphi'(t) \, dt}}$$



证明: $\boxed{F(x)} = \int_a^x f(t) dt \quad x \in I.$

α β

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\begin{aligned} d\boxed{F(\varphi(t))} &= F'(\varphi(t)) \varphi'(t) dt \\ &= \boxed{f(\varphi(t)) \varphi'(t) dt} \end{aligned}$$

Is 2.4k. $\rightarrow$

$$\begin{aligned} \int_a^b f(\varphi(t)) \cdot \varphi'(t) dt &= F \circ \varphi(t) \Big|_{t=a}^{t=b} \\ &= F \circ \underline{\varphi(b)} - F \circ \underline{\varphi(a)} \\ &= F(b) - F(a) \\ &= \int_a^b f(x) dx \end{aligned}$$

例: $\int_0^{\frac{\pi}{2}} \cos^m x dx = \int_0^{\frac{\pi}{2}} \sin^m x dx, \quad m \geq 1$

Ex 13.1:

$$\frac{x = \frac{\pi}{2} - t}{x = \frac{\pi}{2}} \quad dx = -dt \quad t = 0$$

$$\frac{x = 0}{x = 0} \quad t = \frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^m x \, dx = - \int_{\frac{\pi}{2}}^0 \sin^m \left(\frac{\pi}{2} - t \right) dt$$

$$= \int_0^{\frac{\pi}{2}} \sin^m \left(\frac{\pi}{2} - t \right) dt$$

$$= \int_0^{\frac{\pi}{2}} \cos^m t \, dt$$

Ex 13.1: $\int_0^{\pi} \sqrt{1 - \sin x} \, dx$

$$\sin x = \sin 2 \cdot \frac{x}{2}$$

$$= 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$= \int_0^{\pi} \sqrt{1 - 2 \sin \frac{x}{2} \cos \frac{x}{2}} \, dx$$

$$\underbrace{\quad}_{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}}$$

$$= \int_0^{\pi} \sqrt{(\sin \frac{x}{2} - \cos \frac{x}{2})^2} \, dx$$

$$x \in [0, \frac{\pi}{2}] \quad \frac{x}{2} \in [0, \frac{\pi}{4}]$$

$$\sin \frac{x}{2} \leq \cos \frac{x}{2}$$

$$= \int_0^{\frac{\pi}{2}} |\sin \frac{x}{2} - \cos \frac{x}{2}| dx$$

$$= \int_0^{\frac{\pi}{2}} (\cos \frac{x}{2} - \sin \frac{x}{2}) dx + \int_{\frac{\pi}{2}}^{\pi} (\sin \frac{x}{2} - \cos \frac{x}{2}) dx$$

$$x \in [\frac{\pi}{2}, \pi]$$

$$\cos \frac{x}{2} \geq \sin \frac{x}{2}$$

$$= 2(\sin \frac{x}{2} + \cos \frac{x}{2}) \Big|_0^{\frac{\pi}{2}} + 2(-\cos \frac{x}{2} - \sin \frac{x}{2}) \Big|_{\frac{\pi}{2}}^{\pi}$$

$$= 4\sqrt{2} - 1$$

例: $\int_0^a \frac{1}{x + \sqrt{a^2 - x^2}} dx \quad (a > 0)$

$$x = a \sinh t \quad dx = a \cosh t dt$$

$$x=a \Rightarrow t = \frac{\pi}{2}$$

$$x=0 \Rightarrow t=0$$

$$(\sinh t + \cosh t)' = \cosh t - \sinh t$$

$$= \int_0^{\frac{\pi}{2}} \frac{a \cosh t}{a \sinh t + a \cosh t} dt = \int_0^{\frac{\pi}{2}} \frac{\cosh t}{\sinh t + \cosh t} dt$$

$$= \int_0^{\frac{\pi}{2}} (\cosh t - \sinh t) + \sinh t + \cosh t$$

$$-\frac{1}{2} \Big|_0^{\frac{\pi}{2}} \left(\sin t + \cos t \right) dt$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[1 + \frac{(\sin t + \cos t)'}{\sin t + \cos t} \right] dt$$

$$= \frac{\pi}{4} + \frac{1}{2} \ln(\sin t + \cos t) \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{4}$$

例. 5.4.10. 设 $f(x)$ 为连续函数, $F(x) = \int_0^{x^2} t f(x^2 - t) dt$, 求 $F'(x)$.

解:

$$\begin{cases} u = x^2 - t \\ t = x^2 - u \end{cases}$$

$$du = -dt$$

$$t = x^2 \quad u = 0$$

$$t = 0 \quad u = x^2$$

$$F(x) = \int_{x^2}^0 (x^2 - u) f(u) (-du)$$

$$\begin{aligned} G(x) &= \int_0^{x^2} x f(t) dt \\ &= x \int_0^{x^2} f(t) dt \end{aligned}$$

$$= \int_0^{x^2} (x^2 - u) f(u) du$$

$$= \boxed{x^2 \int_0^{x^2} f(u) du} - \boxed{\int_0^{x^2} u f(u) du}$$

$$F'(x) = 2x \int_0^{x^2} f(u) du + \underbrace{x^2 f(x^2) \cdot 2x} - \underbrace{x^2 f(x^2) \cdot 2x}$$

$$= \underline{\underline{2x \int_0^{x^2} f(u) du}}$$

证毕

$$f(x) = f(x + T). \quad (-\infty, +\infty)$$

例 5.4.11. 设 $f(x)$ 是周期为 T 的连续函数, $a \in \mathbb{R}$.

$$\text{证明: } \int_a^{a+T} f(x) dx = \underline{\underline{\int_0^T f(x) dx}}$$

证明:

$$\underline{\underline{x = t - a}}$$

$$\int_a^{a+T} f(x) dx = \boxed{\int_a^0 f(x) dx} + \int_0^T f(x) dx + \int_T^{T+a} f(x) dx$$

$$x = T + t$$

$$dx = dt$$

$$T+a \rightarrow$$

$$a$$

$$T$$

$$0.$$

$$\int_T^{T+a} f(x) dx = \int_0^a f(t) dt = \int_0^a f(x) dx$$

例 5.4.12. 设 $f(x)$ 在 $[a, b]$ 上连续.

$2B$

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx = \frac{1}{2} \int_a^b [f(x) + f(a+b-x)] dx$$

$$= \int_a^{\frac{a+b}{2}} [f(x) + f(a+b-x)] dx$$

$$= \int_{\frac{a+b}{2}}^b \dots$$

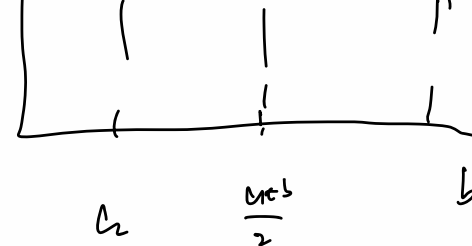
$f(a+b-x)$

$f(x)$

证明

$$\boxed{x = a+b-t} \quad dx = -dt$$

$\begin{matrix} a & b \\ b & a \end{matrix}$



$$\int_a^b f(a+b-x) dx = \int_b^a f(t) (-dt) \\ = \int_a^b f(t) dt$$

$$\begin{matrix} \frac{a+b}{2} & b \\ a & \frac{a+b}{2} \end{matrix}$$

$$\underline{\underline{x = a+b-t}}$$

$$\int_a^{\frac{a+b}{2}} [f(x) + f(a+b-x)] dt = \int_{\frac{a+b}{2}}^b [f(a+b-t) + f(t)] dt \\ = \int_{\frac{a+b}{2}}^b [f(t) + f(a+b-t)] dt$$

推论: 设 $f(x)$ 在 $[-L, L]$ 上连续,

$$\int_{-L}^L f(x) dx = \begin{cases} 0 & f(x) \text{ 奇} \\ 2 \int_0^L f(x) dx & f(x) \text{ 偶} \end{cases}$$

$f(x)$ 奇

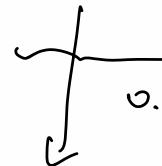
$f(x)$ 偶

$$2 \int_0^L f(x) dx$$

$$\int_a^b f(x) dx = \frac{1}{2} \int_a^b [f(x) + f(a+b-x)] dx$$

证明:

$$\int_{-L}^L f(x) dx = \frac{1}{2} \int_{-L}^L [f(x) + f(-x)] dx$$



推论: 设 $f(x)$ 为奇函数

$$\left\{ \begin{aligned} \int_0^{\frac{\pi}{2}} f(\sin x) dx &= \int_0^{\frac{\pi}{2}} f(\cos x) dx \\ \int_0^{\pi} x f(\sin x) dx &= \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx \end{aligned} \right.$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} f(\sin x) dx &= \int_0^{\frac{\pi}{2}} f(\sin(\frac{\pi}{2}-x)) dx \\ &= \int_0^{\frac{\pi}{2}} f(\cos x) dx \end{aligned}$$

$$\int_a^b f(x) dx$$

$$= \int_a^b f(a+b-x) dx$$

$$x f(\sin x)$$

$$\pi - x$$

$$\begin{aligned}
 \int_0^{\pi} x f(\sin x) dx &= \frac{1}{2} \int_0^{\pi} \left[\underline{x f(\sin x)} + \underline{(\pi-x) f(\sin(\pi-x))} \right] dx \\
 &= \frac{1}{2} \int_0^{\pi} \left[\underline{x f(\sin x)} + \underline{(\pi-x) f(\sin x)} \right] dx \\
 &= \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx
 \end{aligned}$$

$\int_0^{n\pi} x |\sin x| dx$

$|\sin x|$ 周期为 π .

$\int_a^b f(x) dx = \frac{1}{n} \int_a^b [f(x) + f(n\pi - x)] dx$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{n\pi} \left[x |\sin x| + (n\pi - x) |\sin(n\pi - x)| \right] dx \\
 &\quad |\sin(-x)| = |\sin x|
 \end{aligned}$$

$$= \frac{1}{2} \int_0^{n\pi} \left[x \cancel{|\sin x|} + (n\pi - x) \cancel{|\sin x|} \right] dx$$

$n \pi$ 为常数.

$$= \frac{n\pi}{2} \int_0^{n\pi} |\sin x| dx$$

$$= \frac{n\pi}{2} \cdot n \int_0^{\pi} \sin x dx$$

2

$$= n^2 \pi.$$

§ 反常积分.

定义积分.

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

$\left\{ \begin{array}{l} [a, b] \text{ 是有界闭区间} \\ f(x) \text{ 可积} \Rightarrow f(x) \text{ 有界函数.} \end{array} \right.$

① 积分区间无界

无穷积分,

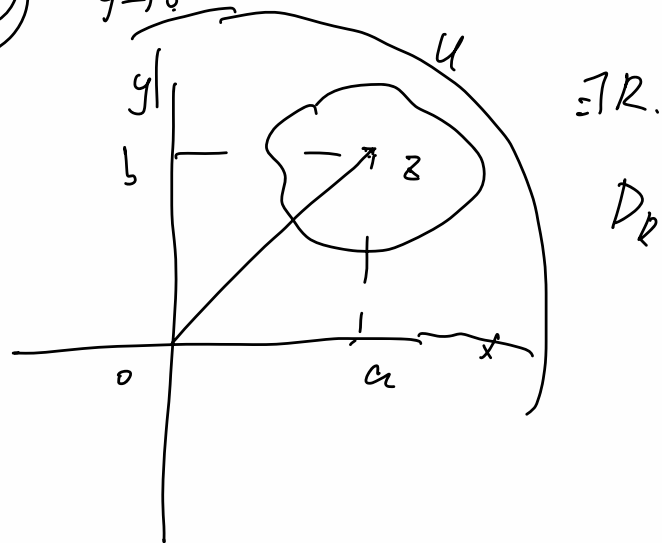
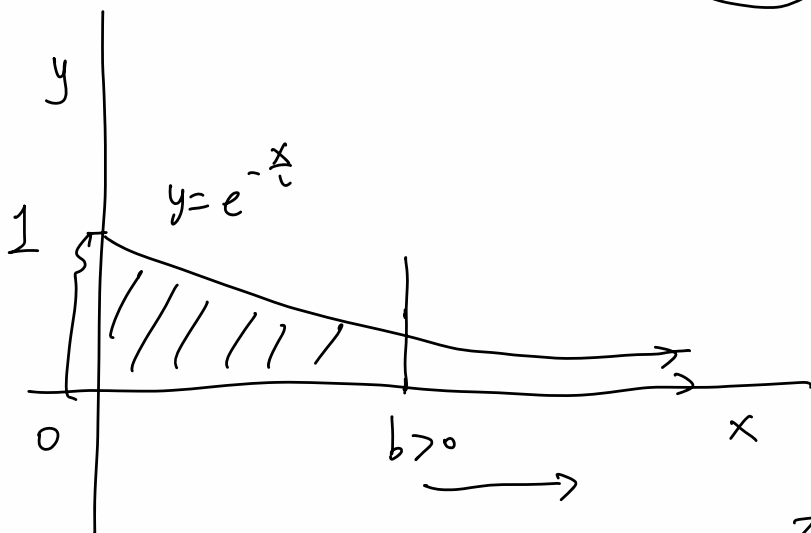
② 无界函数的积分.

例: $y = e^{-\frac{x}{2}}$

$(-\infty, +\infty)$

$\begin{matrix} x \rightarrow -\infty \\ y \rightarrow +\infty \end{matrix}$

$\begin{matrix} x \rightarrow +\infty \\ y \rightarrow 0 \end{matrix}$



$z = (a, b)$

$\sqrt{a^2 + b^2}$

$$A(b) = \int_0^b e^{-\frac{x}{2}} dx$$

$$= -2e^{-\frac{x}{2}} \Big|_0^b = 2 - 2e^{-\frac{b}{2}} \quad 0 < A(b) \uparrow$$

$$\lim_{b \rightarrow +\infty} A(b) = A = \lim_{b \rightarrow +\infty} (2 - 2e^{-\frac{b}{2}})$$

$$= 2$$

$[a, +\infty)$ $(-\infty, b]$ $(-\infty, +\infty)$

定义 5.5.1.

设 $f(x)$ 在 $[a, +\infty)$ 上有定义, 并且对 $\forall b > a$, $f(x)$ 在 $[a, b]$ 上都是可积的.

则称 $f(x)$ 在 $[a, +\infty)$ 上可积.

$$\int_a^b f(x) dx$$

$$b \in [a, +\infty)$$

$$\lim_{b \rightarrow +\infty} \int_a^b f(x) dx \quad \text{存在且有限} \quad \text{称 } f(x) \text{ 收敛.}$$

||

记作 $\int_a^{+\infty} f(x) dx$

不收敛 无穷级数发散.

$$\lim_{a \rightarrow -\infty} \int_a^b f(x) dx = \int_{-\infty}^b f(x) dx$$

设 $f(x)$ 在 $(-\infty, +\infty)$ 上有定义. $\forall b > a$, 设 $f(x)$ 在 $[a, b] \subseteq \mathbb{R}$ 上可积.

任取 $c \in (-\infty, +\infty)$, c 和 a, b 无关.

$$(-\infty, c) \cup (c, +\infty)$$

$$\int_{-\infty}^{+\infty} f(x) dx =$$

$$\int_{-\infty}^c f(x) dx + \int_c^{+\infty} f(x) dx$$

||

$$\lim_{a \rightarrow -\infty} \int_a^c f(x) dx$$

$$\lim_{b \rightarrow +\infty} \int_c^b f(x) dx$$

命题 5.5.2. (线性性质).

若 $\int_a^{+\infty} f(x) dx$, $\int_a^{+\infty} g(x) dx$ 都收敛, 则

$$\int_a^{+\infty} (k_1 f(x) + k_2 g(x)) dx$$

$$= k_1 \int_a^{+\infty} f(x) dx + k_2 \int_a^{+\infty} g(x) dx.$$

k_1, k_2 为常数.

命题 5.5.3. (区间可加性) $x \neq +\infty$,

$f(x)$ 在 $[a, +\infty)$ 上有定义, $[a, u)$ 可积的, $\forall b > a$,

$$\int_a^{+\infty} f(x) dx \text{ 和 } \int_b^{+\infty} f(x) dx \text{ 同敛散.}$$

$$\int_a^{+\infty} f(x) dx = \int_a^b f(x) dx + \int_b^{+\infty} f(x) dx$$

例 5.5.1. $\int_a^{+\infty} \frac{1}{x^\alpha} dx \quad (\alpha > 0).$

$$= \lim_{b \rightarrow +\infty} \int_a^b \frac{1}{x^\alpha} dx$$

$$\alpha = 1 \quad \int_a^b \frac{1}{x} dx = \ln x \Big|_a^b = \ln b - \ln a.$$

$$\lim_{b \rightarrow +\infty} (\ln b - \ln a) = +\infty. \quad \Rightarrow \quad \zeta \text{ diverges.}$$

$$\alpha \neq 1 \quad \int_a^b \frac{1}{x^\alpha} dx = \frac{1}{1-\alpha} x^{1-\alpha} \Big|_a^b$$

$$= \frac{1}{1-\alpha} [b^{1-\alpha} - a^{1-\alpha}] = \frac{1}{\alpha-1} a^{1-\alpha}.$$

$$\frac{1}{2} \alpha < 1 \quad 1-\alpha > 0, \quad b^{1-\alpha} \rightarrow +\infty, \quad \int_a^{+\infty} \frac{1}{x^\alpha} dx \quad \zeta \text{ diverges.}$$

$$\frac{1}{2} \alpha > 1 \quad 1-\alpha < 0, \quad b^{1-\alpha} \rightarrow 0, \quad \int_a^{+\infty} \frac{1}{x^\alpha} dx = \frac{1}{\alpha-1} a^{1-\alpha}.$$

Ex 5.5.2. $\int_2^{+\infty} \frac{1}{x \ln^p x} dx$

($b > 2$)

$$= \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x \ln^p x} dx$$

$$= \lim_{b \rightarrow \infty} \int_2^b \frac{1}{\ln^p x} d \frac{1}{\ln x}$$

$$\frac{1}{2} p=1 \text{ 時 } \int_2^b \frac{1}{\ln x} d \frac{1}{\ln x} = \ln \ln x \Big|_2^b = \ln \ln b - \ln \ln 2$$

$\lim_{b \rightarrow \infty} \ln b \rightarrow \infty$
 $\lim_{b \rightarrow \infty} \ln \ln b \rightarrow \infty$

4. 2.

$$\int \frac{1}{x^p} dx$$

$$\frac{1}{2} p \neq 1 \text{ 時 } \int_2^b \frac{1}{\ln^p x} d \frac{1}{\ln x} = \frac{1}{1-p} \ln^{1-p} x \Big|_2^b$$

$$= \frac{1}{1-p} [\ln^{1-p} b - \ln^{1-p} 2]$$

$p < 1$ 時

$$\ln^{1-p} b \rightarrow \infty$$

$p > 1$ 時

$$\ln^{1-p} b \rightarrow 0$$

$$\int_2^{\infty} \frac{1}{x \ln^p x} dx$$

$$= \frac{1}{p-1} \ln^{1-p} 2$$

Newton - Leibniz 則.

$$\text{Def. } f(x), \quad F(x) \text{ is } f(x)\text{-antiderivative,} \quad \begin{aligned} F(+\infty) &= \lim_{x \rightarrow +\infty} F(x), \\ F(-\infty) &= \lim_{x \rightarrow -\infty} F(x) \end{aligned}$$

$f(x)$ is on $[a, +\infty)$ or $(-\infty, b]$.

$$\left\{ \begin{aligned} \int_a^{+\infty} f(x) dx &= F(+\infty) - F(a) = F(x) \Big|_a^{+\infty} \\ \int_{-\infty}^b f(x) dx &= F(b) - F(-\infty) = F(x) \Big|_{-\infty}^b \\ \int_{-\infty}^{+\infty} f(x) dx &= F(+\infty) - F(-\infty) = F(x) \Big|_{-\infty}^{+\infty} \end{aligned} \right\}$$

Ex. 5.5.4. $\int_{\frac{2}{\pi}}^{+\infty} \frac{1}{x^2} \sin \frac{1}{x} dx$

$$= \int_{\frac{2}{\pi}}^{+\infty} -\sin \frac{1}{x} d \frac{1}{x}$$

$$= \cos \frac{1}{x} \Big|_{\frac{2}{\pi}}^{+\infty}$$

$$= \lim_{x \rightarrow +\infty} \cos \frac{1}{x} - \cos \frac{2}{v}$$

$$= 1.$$

例: $\int_2^{+\infty} \frac{dx}{x^2+x-2}$

$$= \int_2^{+\infty} \frac{1}{(x+2)(x-1)} dx$$

$$= \frac{1}{3} \int_2^{+\infty} \left(\frac{1}{x-1} - \frac{1}{x+2} \right) dx$$

$$= \frac{1}{3} \left[\ln(x-1) - \ln(x+2) \right] \Big|_2^{+\infty}$$

$$= \frac{1}{3} \ln \left(\frac{x-1}{x+2} \right) \Big|_2^{+\infty}$$

$$= \int_2^{+\infty} \frac{1}{x-1} dx - \int_2^{+\infty} \frac{1}{x+2} dx$$

$$= \left[\ln(x-1) \right]_2^{+\infty} - \left[\ln(x+2) \right]_2^{+\infty}$$

$$= -\frac{1}{3} \ln \frac{1}{4} = \frac{1}{3} \ln 4 = \frac{2}{3} \ln 2.$$

Ex 1: $a > 0, \int_0^{+\infty} e^{-ax} \cos bx \, dx \quad \& \quad \int_0^{+\infty} e^{-ax} \sin bx \, dx.$

$$= \frac{1}{b} \int_0^{+\infty} e^{-ax} d(\sin bx)$$

$$= \frac{1}{b} \left\{ \underbrace{e^{-ax} \sin bx \Big|_0^{+\infty}}_{\substack{\downarrow 0 \\ \downarrow \frac{1}{b}}} - \int_0^{+\infty} \sin bx \, d(e^{-ax}) \right\}$$

$$= \frac{a}{b} \int_0^{+\infty} \sin bx \, e^{-ax} \, dx = \frac{b}{a^2 + b^2}$$

$$= -\frac{a}{b^2} \int_0^{+\infty} e^{-ax} d(\cos bx)$$

$$= -\frac{a}{b^2} \left\{ \underbrace{e^{-ax} \cos bx \Big|_0^{+\infty}}_{0-1=-1} - \int_0^{+\infty} \cos bx \, d(e^{-ax}) \right\}$$

$$= \frac{a}{b^2} \int_0^{+\infty} e^{-ax} \cos bx \, dx$$

$$= \frac{a}{b^2} - \frac{a^2}{b^2} \int_0^{+\infty} e^{-ax} \cos bx \, dx$$

$$I = \frac{a}{b^2} - \frac{a^2}{b^2} I$$

$$I = \frac{a}{b^2} \cdot \frac{b^2}{a^2 + b^2} = \frac{a}{a^2 + b^2}$$

例:

$$\int_0^{+\infty} \frac{1}{(a^2 + x^2)^{\frac{3}{2}}} dx$$

设 $a > 0$.

$$x = a \tan t$$

$$dx = \frac{a}{\cos^2 t} dt$$

$$x = 0 \quad t = 0$$

$$x = +\infty \quad t = \frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{(a^2 \sec^2 t)^{\frac{3}{2}}} \cdot \frac{a}{\cos^2 t} dt$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{a \sec^2 t} dt$$

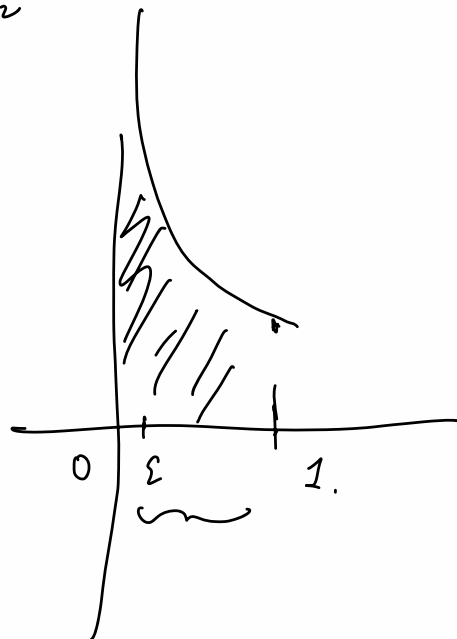
$$= \int_0^{\frac{\pi}{2}} \frac{1}{a^3 \cdot \sec^2 t} dt$$

$$= \frac{1}{a^2} \int_0^{\frac{\pi}{2}} \left(\frac{1}{\sec t} \right)^{\cos t} dt = \frac{1}{a^2}$$

无界区间的反常积分.

例: $\int_0^1 \frac{1}{\sqrt{x}} dx$

$$\lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}} = +\infty.$$



$$\forall \varepsilon \in (0, 1) \quad t_\varepsilon \in [\varepsilon, 1].$$

$$\int_\varepsilon^1 \frac{1}{\sqrt{x}} dx = 2\sqrt{x} \Big|_\varepsilon^1 = 2 - 2\sqrt{\varepsilon}.$$

$$\lim_{\varepsilon \rightarrow 0^+} \int_\varepsilon^1 \frac{1}{\sqrt{x}} dx = 2.$$

收敛

$$\left\{ \begin{array}{l} \frac{[a, b)}{[a, b]} \\ [a, b] \ni c. \quad (a, b). \end{array} \right.$$

① 设 $f(x)$ 在 $[a, b]$ 上^右连续, $\forall \varepsilon > 0$, $f(x)$ 在 $[a, b-\varepsilon]$ 上可积.

$$\lim_{x \rightarrow b^-} f(x) = m. \quad (\text{右极限})$$

$$\lim_{\varepsilon \rightarrow 0^+} \int_a^{b-\varepsilon} f(x) dx$$

称 $f(x)$ 在 $[a, b]$ 上反常积分
收敛

$$\text{记 } \int_a^b f(x) dx.$$

② $[a, b]$. $\forall \varepsilon > 0$, $f(x)$ 在 $[a-\varepsilon, b]$ 上可积. $\lim_{x \rightarrow a^+} f(x) = m.$

$$\int_a^b f(x) = \lim_{\varepsilon \rightarrow 0^+} \int_{\underline{a+\varepsilon}}^b f(x) dx$$

③ $f(x)$ 在 $[a, b]$ 内有瑕点 $c \in [a, b]$. $[a, c)$ 及 $(c, b]$

$$\underline{\underline{\int_a^b f(x) dx = \underbrace{\int_a^{\textcircled{c}} f(x) dx}_{\text{I.}} + \underbrace{\int_{\textcircled{c}}^b f(x) dx}_{\text{II.}}}}$$

④ (a, b) . $a, b \in \mathbb{R}$ $\forall c \in (a, b)$. $(a, c] \cup [c, b)$

$$\underline{\underline{\int_a^b f(x) dx = \underbrace{\int_a^c f(x) dx}_{\text{I.}} + \underbrace{\int_c^b f(x) dx}_{\text{II.}}}}$$

Ex. 1: $\int_0^1 \frac{1}{x^2} dx$ ($x > 0$)

$x=0$ ist ∞ . $\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty$.

$\frac{1}{2} x = 2 \text{ ist.}$

$$\underline{\underline{\int_0^1 \frac{1}{x} dx = \lim_{\varepsilon \rightarrow 0^+} \underbrace{\int_{\varepsilon}^1 \frac{1}{x} dx}_{\downarrow \text{Zusatz.}} = \lim_{\varepsilon \rightarrow 0^+} (\ln x|_{\varepsilon}^1) = \lim_{\varepsilon \rightarrow 0^+} (-\ln \varepsilon) = +\infty.}}$$

$$\frac{1}{2}\alpha \neq 1 \text{ 时} \quad \int_0^1 \frac{1}{x^\alpha} dx = \lim_{\varepsilon \rightarrow 0^+} \int_\varepsilon^1 \frac{1}{x^\alpha} dx$$

$$= \lim_{\varepsilon \rightarrow 0^+} \left(\frac{1}{1-\alpha} - \frac{1}{1-\alpha} \varepsilon^{1-\alpha} \right)$$

$$\frac{1}{2}\alpha < 1 \text{ 时, } \varepsilon^{1-\alpha} \rightarrow 0. \quad \text{收敛} \quad \alpha < 1 \quad \text{收敛}$$

$$\frac{1}{2}\alpha > 1 \text{ 时, } \varepsilon^{1-\alpha} \rightarrow +\infty. \quad \text{发散} \quad \alpha \geq 1 \quad \text{发散}$$

$$\int_1^{+\infty} \frac{1}{x^\alpha} dx \quad \begin{matrix} \alpha > 1 & \text{收敛} \\ \alpha \leq 1 & \text{发散} \end{matrix} \quad \swarrow \quad \underline{\int_0^{+\infty} \frac{1}{x^\alpha} dx}$$

$$\int_{-1}^1 \frac{1}{x^2} dx \quad x=0 \text{ 瑕点}$$

$$= \int_{-1}^0 \frac{1}{x^2} dx + \int_0^1 \frac{1}{x^2} dx$$

$$\int_{-1}^1 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_{-1}^1 = -2 \quad \textcircled{X}$$

$$f(x) \text{ 在 } [a, b] \text{ 上连续. } F'(x) = f(x), \quad \forall x \in (a, b) \text{ 上.}$$

$$\int_a^b f(x) dx = \lim_{x \rightarrow a^+} \underbrace{F(x)}_{\text{右连续 (右极限)}} = \underbrace{F(b) - F(a^+)}_{\substack{\downarrow \\ [a, b]}} = F(x) \Big|_a^b$$

例: 5.5.10.

$$\int_0^1 \frac{1}{\sqrt{x}(1+x)} dx \quad \underline{\underline{x=0 \text{ 瑕点.}}}$$

$$\text{令 } t = \sqrt{x} \quad x = t^2 \quad dx = 2t dt.$$

$$x=1 \quad t=1$$

$$x=0 \quad t=0.$$

$$\int_0^1 \frac{1}{t(1+t^2)} \cdot 2t dt = \int_0^1 \frac{2}{1+t^2} dt.$$

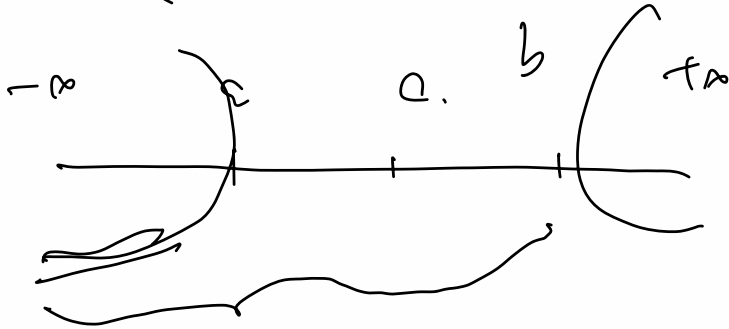
$$= 2 \int_0^1 \frac{1}{1+t^2} dt$$

$$= 2 \underbrace{\arctan t \Big|_0^1}_{\frac{\pi}{4} - 0} = \left(\frac{\pi}{2} \right)$$

C.

$$\int_{-\infty}^{+\infty} f(x) dx$$

$(-\infty, +\infty)$



$$= \underbrace{\int_{-\infty}^a + \int_a^b + \int_b^{+\infty}}$$

$$e^{-t^2}$$

12.1. $f(x) = \int_1^{\sqrt{x}} e^{-t^2} dt$

$$\int_0^1 \frac{f(x)}{\sqrt{x}} dx$$

$x=0$ {illegible}

$$\int_0^1 \frac{f(x)}{\sqrt{x}} dx = 2 \int_0^1 f(x) d\sqrt{x}$$

$$= \left(2 f(x) \sqrt{x} \Big|_0^1 \right) - 2 \int_0^1 \sqrt{x} df(x)$$

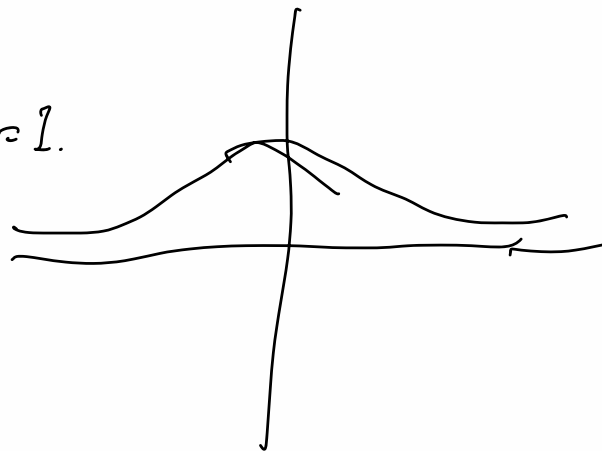
$$= -x \int_0^1 \cancel{dx} e^{-x} \cdot \frac{1}{\cancel{dx}} dx$$

$$= - \int_0^1 e^{-x} dx$$

$$= e^{-x} \Big|_0^1 = \frac{1}{e} - 1.$$

$$\underline{\underline{f(x) = e^{-x^2}}}$$

$$x \rightarrow \infty, f(x) = 1.$$



$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$\begin{matrix} x \\ \downarrow \\ [a, b] \end{matrix}$$

$$\int_a^b f(x) dx$$

$$\int_{-\infty}^{+\infty} f(x) dx = 1$$

$$\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx$$

$$= \sqrt{2\pi}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx$$

$$\sqrt{2\pi}$$

$$\int_0^{+\infty} e^{-\frac{x}{2}} dx = \frac{1}{2}$$

$$\int_x^{\infty} e^{-t^2} dx$$

$f(t)$ 有界

例: $\lim_{x \rightarrow 0+} x^{\alpha} \int_x^1 \frac{f(t)}{t^{\alpha+1}} dt, \quad \alpha > 0, \quad f(x) \text{ 在 } [0, 1] \text{ 上连续.}$

$\underbrace{\hspace{10em}}_0$

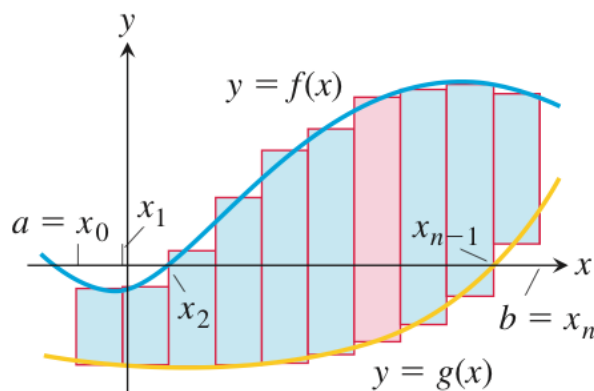
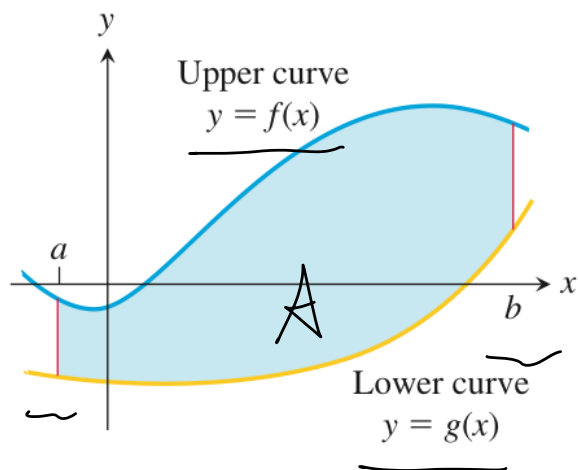
$$= \lim_{x \rightarrow 0+} \frac{\int_x^1 \frac{f(t)}{t^{\alpha+1}} dt}{\frac{1}{x^{\alpha}}} (+\infty)$$

$$\lim_{x \rightarrow 0+} f(x) = f(0)$$

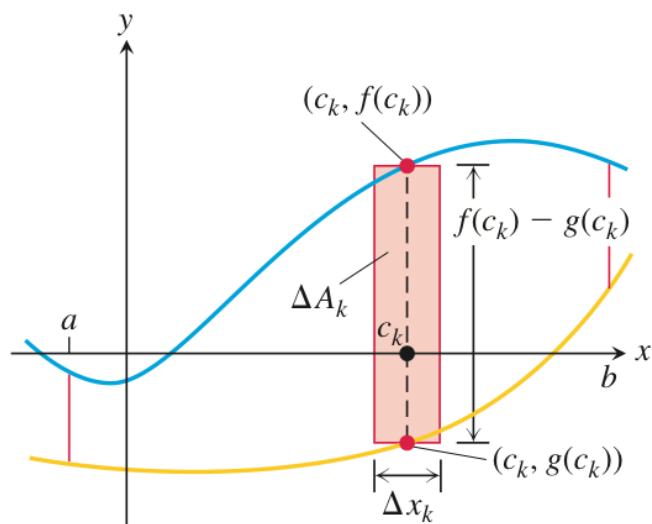
$$= \lim_{x \rightarrow 0+} \frac{f(x) + \frac{f(x)}{x^{\alpha+1}}}{f(x) \cdot \frac{1}{x^{\alpha+1}}} = \lim_{x \rightarrow 0+} \frac{f(x)}{\frac{1}{x^{\alpha}}} = \frac{f(0)}{\frac{1}{x^{\alpha}}}$$

§ 平面图形的面积.

$$f(x) \geq 0. \quad \int_a^b f(x) dx \geq 0.$$



$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b.$$



$$\xi_k \in [x_{k-1}, x_k].$$

$$\sum_{k=1}^n [f(\xi_k) - g(\xi_k)] \Delta x_{k-1}$$

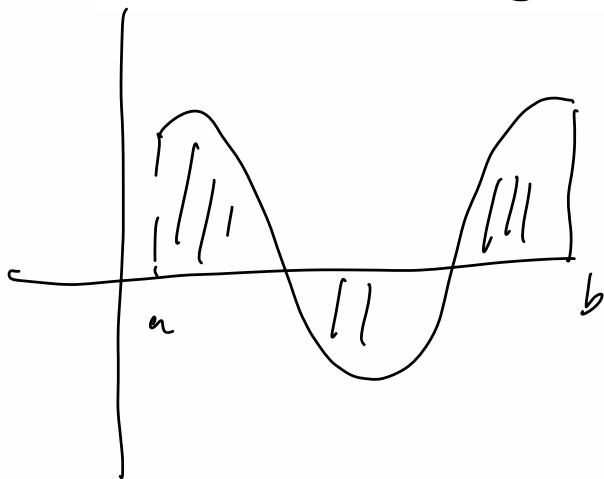
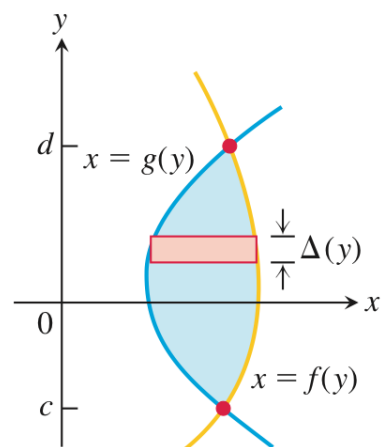
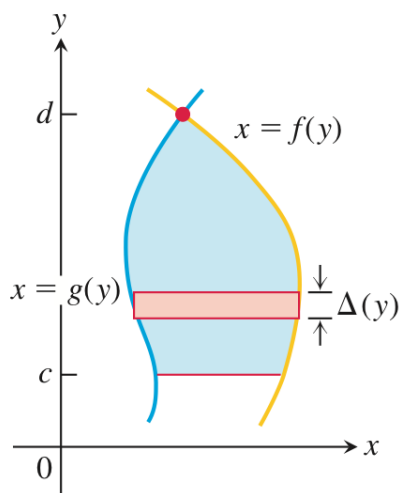
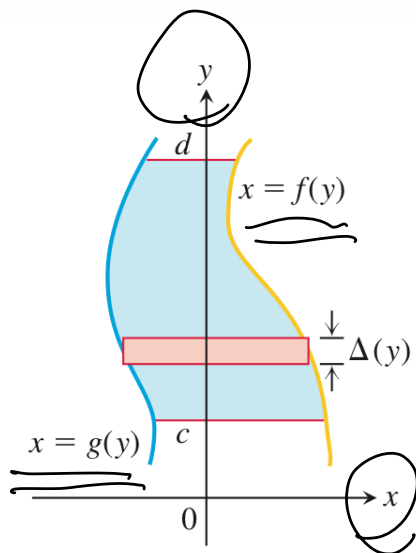
$$(f - g)(\xi_k)$$

$$\int_a^b [f(x) - g(x)] dx.$$

$$\lambda = \max_k \Delta x_k$$

$\lambda \rightarrow 0$

$$\int_c^d [f(y) - g(y)] dy.$$



$$A = \int_a^b |f(x)| dx$$

例 6.1.1 求抛物线 $y^2 = x + 4$ 与直线 $x + y = 2$ 围成图形的面积 A .

① 以 x 为自变量. $y = 2 - x$.

$$\begin{cases} y \geq 0, & y = \sqrt{x+4} \\ y \leq 0, & y = -\sqrt{x+4} \end{cases}$$

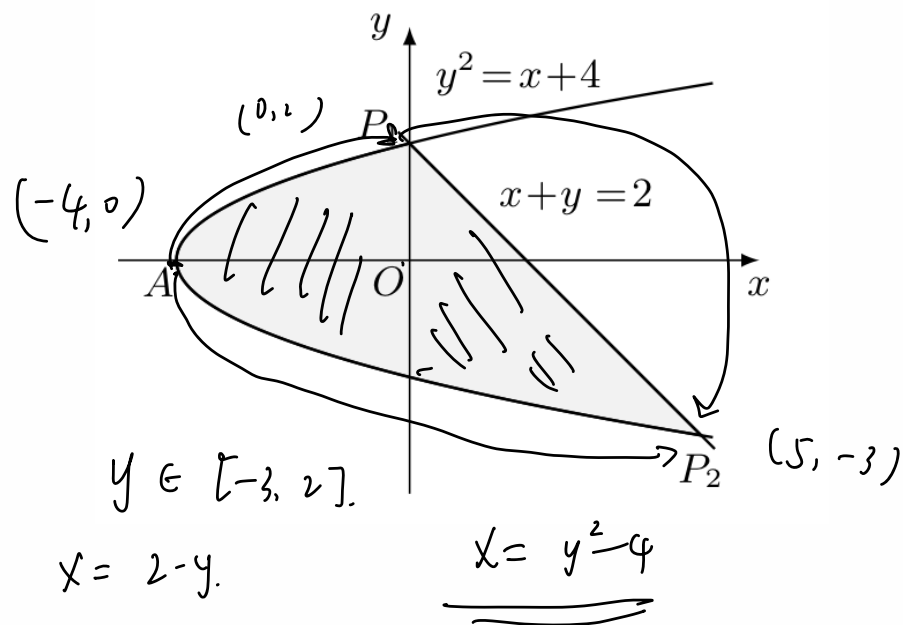
$$\int_{-4}^0 (\sqrt{x+4} - (-\sqrt{x+4})) dx$$

$$+ \int_0^5 [(2-x) - (-\sqrt{x+4})] dx$$

$$= \frac{125}{6}$$

$$\int_{-3}^2 [(2-y) - (y^2-4)] dy.$$

$$= \frac{125}{6}$$



例 6.1.2 如图 6.3, C_1 和 C_2 分别是 $y = \frac{1}{2}(1 + e^x)$ 和 $y = e^x$ 的图像, 过点 $(0, 1)$ 的曲线 C_3 是一条单调增函数的图像. 过 C_2 上任一点 $M(x, y)$ 分别作垂直于 x 轴和 y 轴的直线 l_x 和 l_y . 记 C_1, C_2 与 l_x 所围图形的面积为 $S_1(x)$; C_2, C_3 与 l_y 所围图形的面积为 $S_2(x)$; 如果总有 $S_1(x) = S_2(y)$, 求曲线 C_3 的方程 $x = \varphi(y)$.

$$S_1 = \int_0^x [e^t - \frac{1}{2}(1 + e^t)] dt.$$

$$y \in [1, e^x].$$

$$y = e^x \\ x = \ln y.$$

$$= S_2 = \int_1^{e^x} [\ln y - \varphi(y)] dy.$$

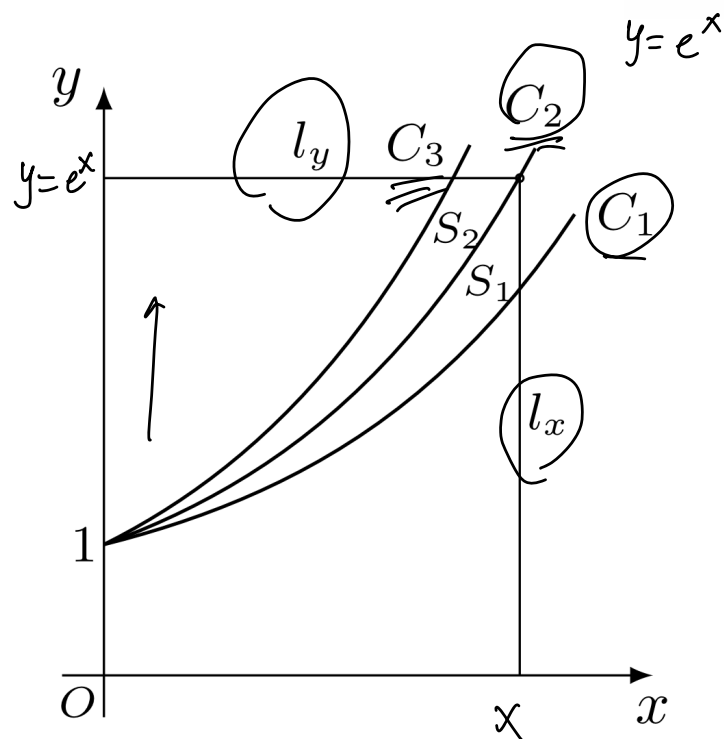
$$\underline{\underline{x = \varphi(y)}}.$$

$$\frac{1}{2} e^x - \frac{1}{2} = [\ln e^x - \underline{\varphi(e^x)}] e^x$$

$$\varphi(e^x) = \ln e^x + \frac{1}{2e^x} - \frac{1}{2}$$

$$y = e^x$$

$$\varphi(y) = \ln y + \frac{1}{2y} - \frac{1}{2}.$$



例 6.1.3 假设函数 $f(x)$ 在区间 $[a, b]$ 上连续, 且在 (a, b) 内有 $f'(x) > 0$. 证明: 在 (a, b) 内存在唯一的 ξ , 使曲线 $y = f(x)$ 与两直线 $y = f(\xi)$, $x = a$ 所围平面图形面积 S_1 是曲线 $y = f(x)$ 与两直线 $y = f(\xi)$, $x = b$ 所围平面图形面积 S_2 的 3 倍.

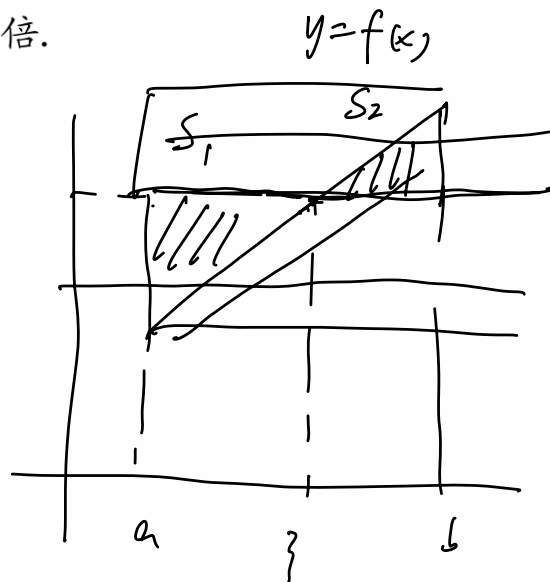
$$S_1 = 3S_2.$$

$$S_1 - 3S_2 = 0.$$

$$S_1 = \int_a^{\xi} (f(\xi) - f(x)) dx$$

$$S_2 = \int_{\xi}^b (f(x) - f(\xi)) dx$$

$t = \xi$



$$F(t) = \int_a^t [f(t) - f(x)] dx - 3 \int_t^b [f(x) - f(t)] dx$$

$$F(a) = -3 \int_a^b (f(x) - f(a)) dx < 0$$

$$F(b) > 0.$$

$$F'(t) = \left[\frac{f(t) \int_a^t dx - \int_a^t f(x) dx}{(t-a)} \right] \ominus \frac{3 \int_t^b f(x) dx}{-3 f(t)} + \frac{f(t) \int_t^b dx}{f(t)}$$

$$f'(t)(t-a) + f(t) - f(t) + 3f(t) + 3f(t)(1-t) = 3f(t)$$

$$= \underline{\underline{f'(t) [\underline{\underline{(t-a)} + 3 \underline{\underline{(b-t)}}]}}}. > 0.$$

例 6.1.4 求星形线 $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$ ($a > b > 0$) 所围成图形的面积 A .

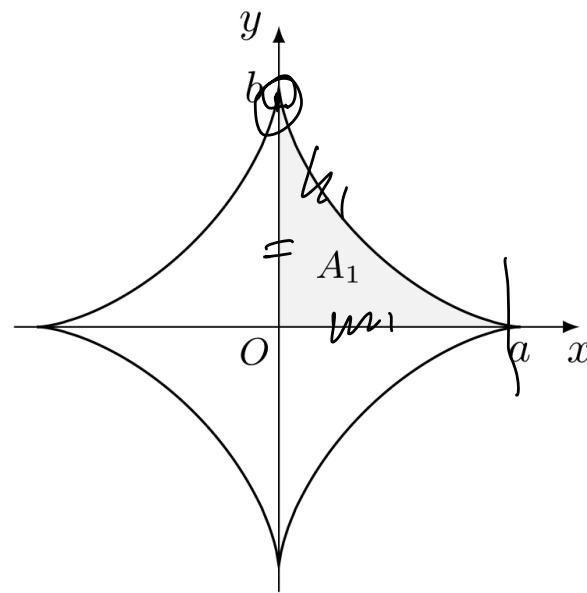
内摆线.

$$y = b \left(1 - \left(\frac{x}{a} \right)^{2/3} \right)^{3/2}.$$

$$\begin{cases} x = a \cos^3 \theta \\ y = b \sin^3 \theta \end{cases} \quad (\alpha=1)$$

$$A = 4A_1 = 4b \int_0^a \left[1 - \left(\frac{x}{a} \right)^{2/3} \right]^{3/2} dx$$

$$= 4b \int_0^a \left[1 - \left(\left(\frac{x}{a} \right)^{1/3} \right)^2 \right]^{3/2} dx$$



$$1 - n^2$$

$$u = \cos \varphi.$$

$$x = a \cos^2 t. \quad t \in [0, \frac{\pi}{2}].$$

$$= \frac{3}{8} \pi.$$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$$[\alpha, \beta]$$

$$\downarrow$$

$$[a, b].$$

$$y = y(x)$$

$$\int_a^b |y| dx$$

$$= \int_a^b (y(t)) \underline{\underline{x'(t) dt}}$$

$$\underline{\underline{x'(t) dt}}$$

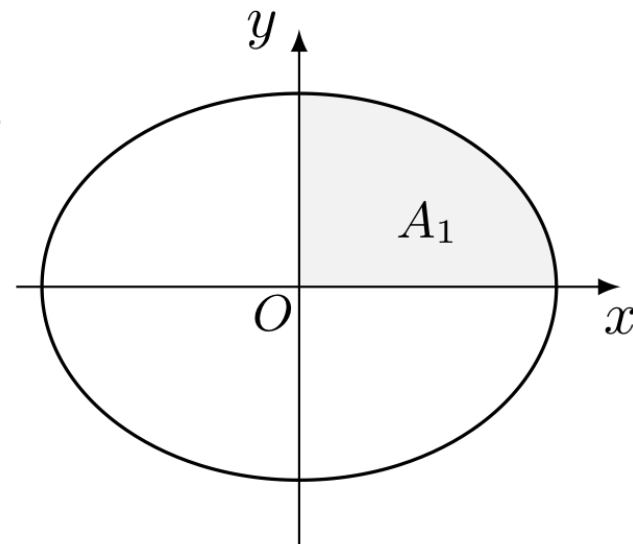
$$a < b$$

$$\alpha > \beta$$

例 6.1.5 求椭圆 $x = a \cos t$, $y = b \sin t$, ($a > 0, b > 0$) 的面积.

$$A_1 = \int_0^{\frac{\pi}{2}} (y \cdot x') dt$$

$$= 4ab \int_0^{\frac{\pi}{2}} \underbrace{\sin^2 t} dt = ab\pi.$$



§ 极坐标.

极径 r

(r, θ)

$r > 0$.

极角 θ .

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \arctan\left(\frac{y}{x}\right) \quad x > 0$$

阿基米德螺线.

$$r = b\theta$$

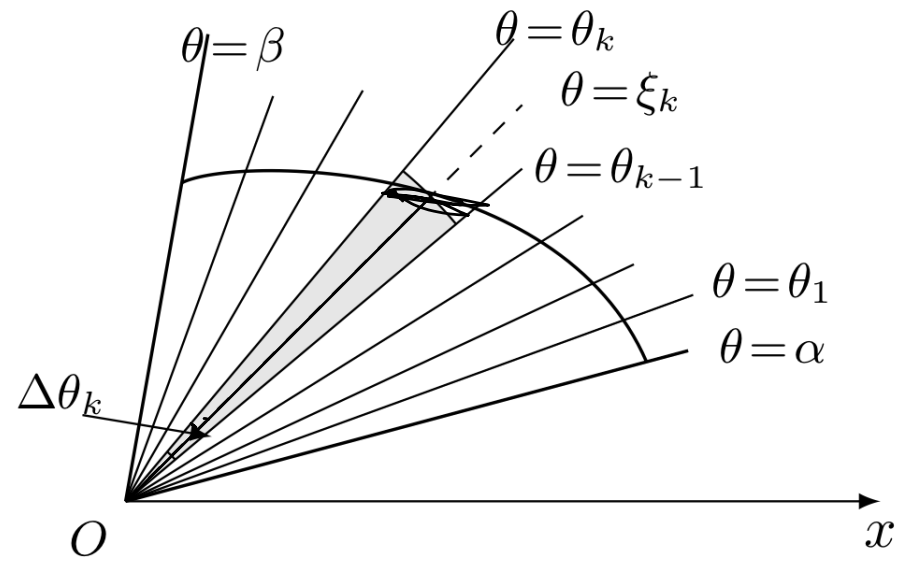
$$\theta \in [0, \pi]$$



$$r(\theta_k),$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{2} r(\theta_k)^2 \Delta \theta_k$$

$$= \int_{\alpha}^{\beta} \frac{1}{2} r^2(\theta) d\theta = A$$



例 6.1.6 求一个动圆绕着一个与其相切且半径相等的定圆外部作无滑动地滚动一周时，动圆上一个固定点走过的轨迹方程。

外摆线。



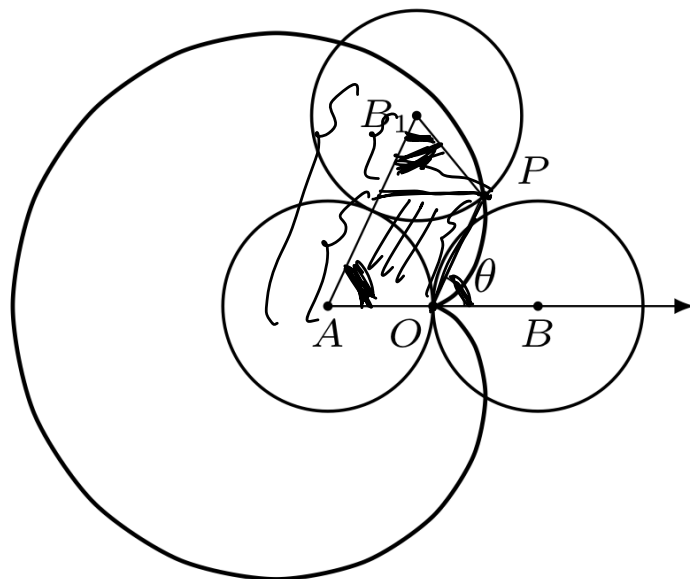
$$2a - 2a \cos \theta$$

$$r = 2a(1 - \cos \theta), \quad \theta \in [0, \pi].$$

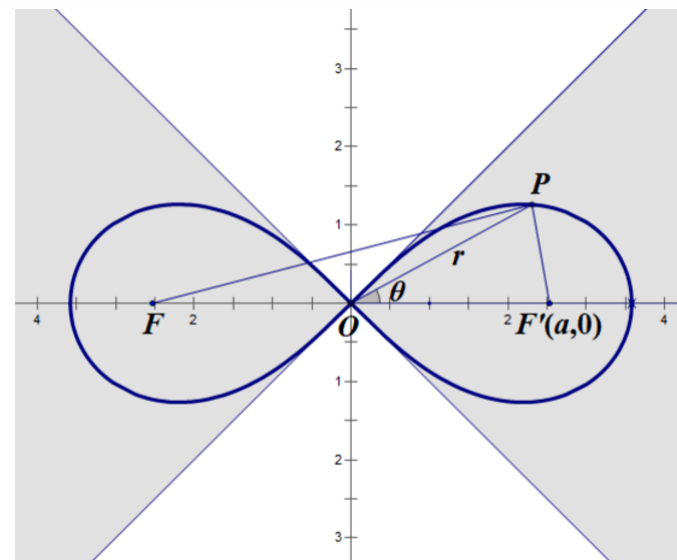
$$A = 2 \int_0^{\pi} \frac{1}{2} r^2(\theta) d\theta$$

$$= 2 \cdot \frac{1}{2} \int_0^{\pi} 4a^2 (1 - \cos \theta)^2 d\theta$$

$$= 6a^2\pi.$$



例：求双纽线 $r^2 = 2a^2 \cos 2\theta$ 所围成平面图形的面积



例: 求 $r = a \sin 3\theta$ 所围成的面积 (三叶玫瑰线)

$$r = a \sin(k\theta) \quad k \geq 1.$$

$k \in \text{奇数}$ $k \neq \frac{1}{2}$

$k \in \text{偶数}$ $2k \neq \frac{1}{2}$

$r \geq 0$.

$$\sin 3\theta \geq 0$$

$$\begin{cases} [0, \frac{\pi}{3}] \\ [\frac{2}{3}\pi, \pi] \\ [\frac{4}{3}\pi, \frac{5}{3}\pi] \end{cases}$$

$$A = 6 \int_0^{\frac{\pi}{6}} \frac{1}{2} r^2 d\theta$$

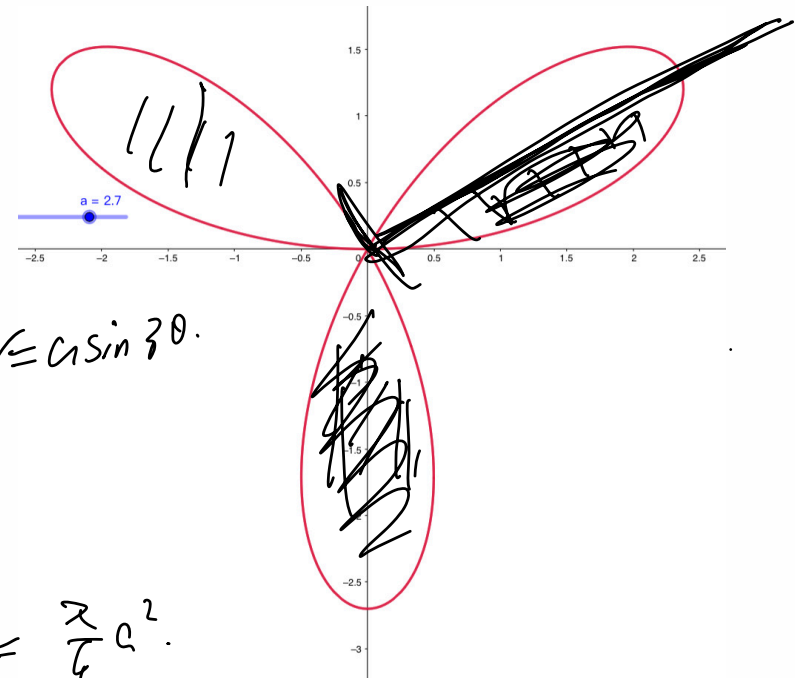
$$= 6 \int_0^{\frac{\pi}{6}} \frac{1}{2} a^2 \sin^2 3\theta d\theta$$

$$= \frac{\pi}{4} a^2.$$

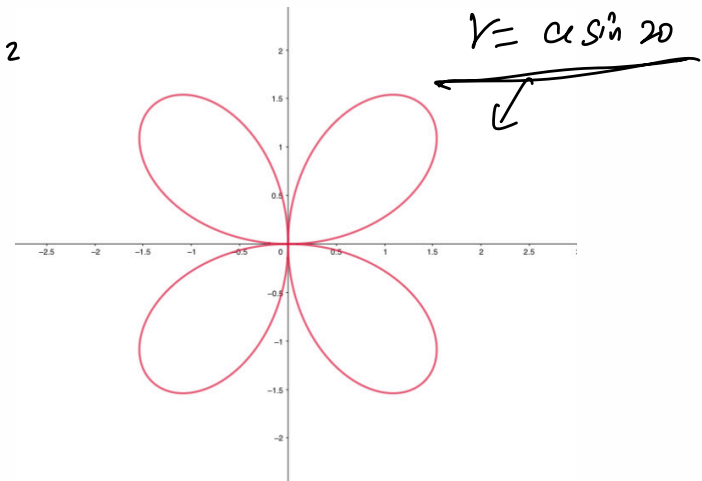
$$[0, \frac{\pi}{6}]$$

$$k \text{ 奇数} \quad A = \frac{\pi}{4} a^2.$$

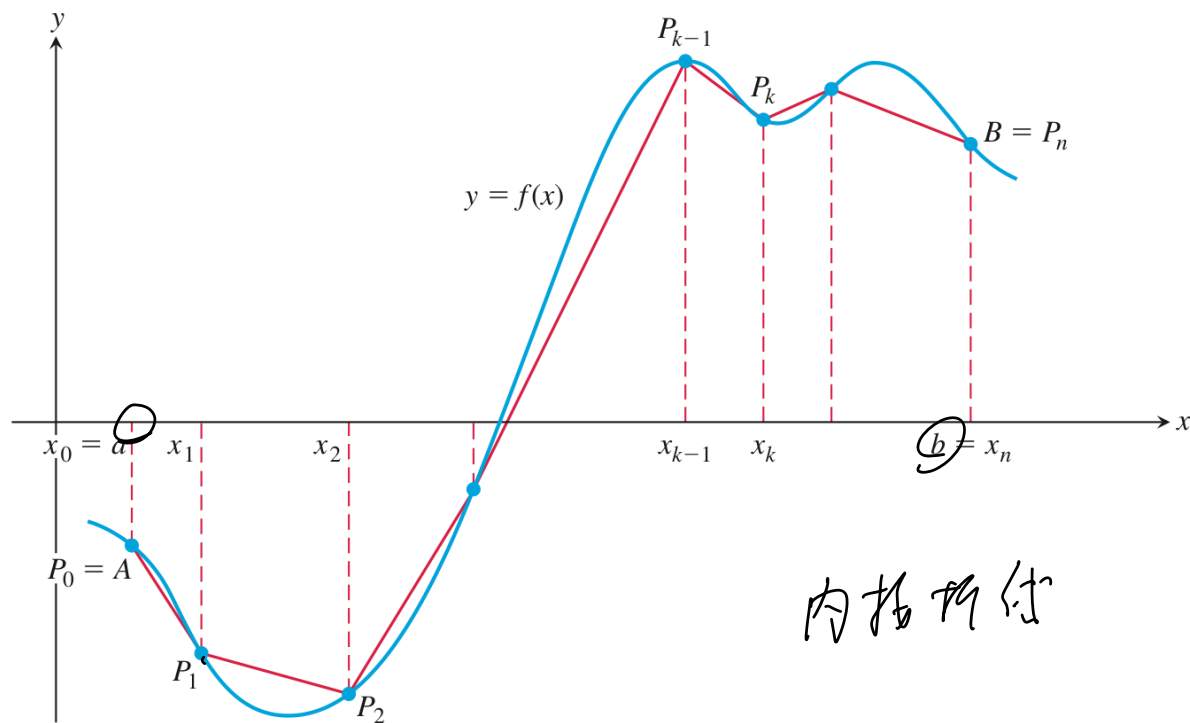
$$k \text{ 偶数} \quad A = \frac{\pi}{2} a^2.$$



$$r = a \sin 3\theta.$$



§ 平面曲线的弧长



内插折线

$$x \in [a, b].$$

$$t \in [\alpha, \beta].$$

$$\alpha = t_0 < t_1 < \dots < t_{n-1} < t_n = \beta.$$

$$\Delta t_i = t_i - t_{i-1}, \quad \Delta = \max \Delta t_i.$$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$$\lim_{\Delta \rightarrow 0} \sum_{i=1}^n \sqrt{\underbrace{(x(t_i) - x(t_{i-1}))^2}_{\text{可求长度}} + \underbrace{(y(t_i) - y(t_{i-1}))^2}_{\text{可求长度}}} = L$$

可求长度

弧长.

$$\lim_{\Delta \rightarrow 0} \sum_{i=1}^n \sqrt{(x'(t_i) \Delta t_i)^2 + (y'(t_i) \Delta t_i)^2} = \lim_{\Delta \rightarrow 0} \sqrt{(x'(t_i) \Delta t_i)^2 + (y'(t_i) \Delta t_i)^2}$$

ξ_i 和 η_i 不同.

$$\sqrt{1 + (y'(t_i) \Delta t_i)^2}$$

(一致连续性)

1.2.1 ①.

$$= \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt$$

$x'(t)$ 和 $y'(t)$ 连续.

$$x'(t)^2 + y'(t)^2 \neq 0.$$

1.2.1 ② $y=f(x), x \in [a, b] \quad \begin{cases} x=x \\ y=f(x) \end{cases}$

$$\underline{\underline{ds. \text{ 弧长元素 } \hat{z}}}$$

$$S = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

1.2.1 ③. $r=r(\theta), \theta \in [\alpha, \beta].$

$$\begin{cases} x=r \cos \theta \\ y=r \sin \theta \end{cases}$$

$$S = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$$

$$= \int_a^b \sqrt{[r'(\theta)]^2 + [r(\theta)]^2} d\theta$$

$$= \int_a^b \sqrt{r^2 + (r')^2} d\theta.$$

例 6.2.2 求心形线 $r = 2a(1 + \cos \theta)$ 的全长, 其中 $a > 0$ 是常数.

$$\begin{aligned} S &= 2 \int_0^{\pi} \sqrt{r^2 + [r'(\theta)]^2} d\theta \\ &= 2 \int_0^{\pi} 2a \sqrt{(1 + \cos \theta)^2 + \sin^2 \theta} d\theta \\ &\quad \begin{array}{c} 2 + 2 \cos \theta \\ 2(1 + \cos \theta) \end{array} \\ &= 4a \int_0^{\pi} \cos \frac{\theta}{2} d\theta \\ &= 16a. \end{aligned}$$

例 6.2.3 求摆线 $\begin{cases} x = 1 - \cos t, \\ y = t - \sin t, \end{cases}$ 一拱 ($0 \leq t \leq 2\pi$) 的弧长 s .

$$S = \int_0^{2\pi} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$= \int_0^{2\pi} \sqrt{\sin^2 t + (1 - \cos t)^2} dt$$

$$= \int_0^{2\pi} 2 \sin \frac{t}{2} dt = 8.$$

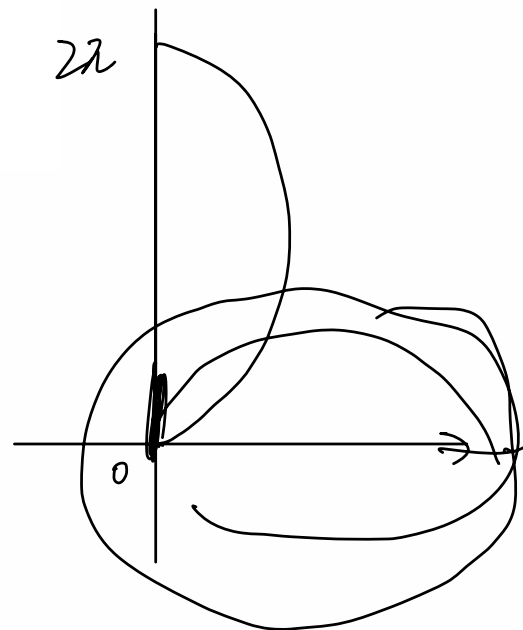
$$dx = \sin t dt$$

$$dy = (1 - \cos t) dt$$

$$\frac{dy}{dx} = \frac{1 - \cos t}{\sin t} = \tan \frac{t}{2}$$

$$t=0 \quad \frac{dy}{dx} = 0.$$

$$\frac{dy}{dx} = \tan,$$



$y = f(x)$, 光滑曲线 可导且导数连续

$x \in [a, b]$

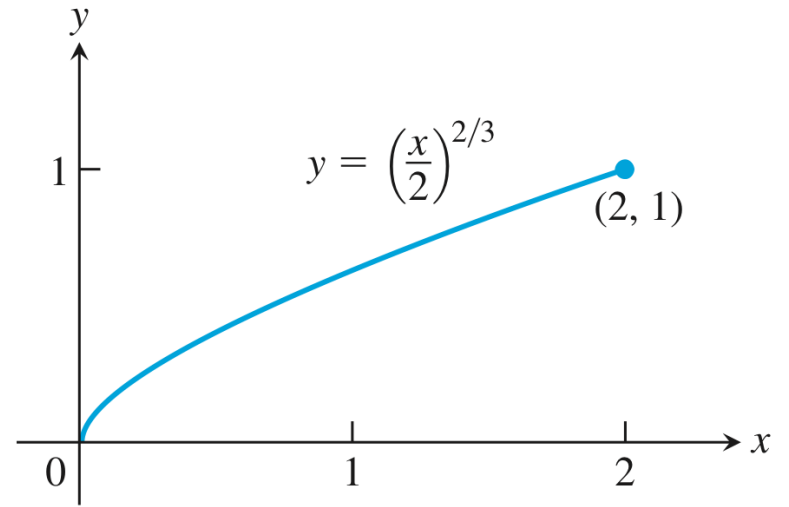
$$\frac{dy}{dx} = f'(x) = \infty.$$

$$x = f^{-1}(y) \quad [f^{-1}(y)]' = \frac{1}{f'(x)}$$

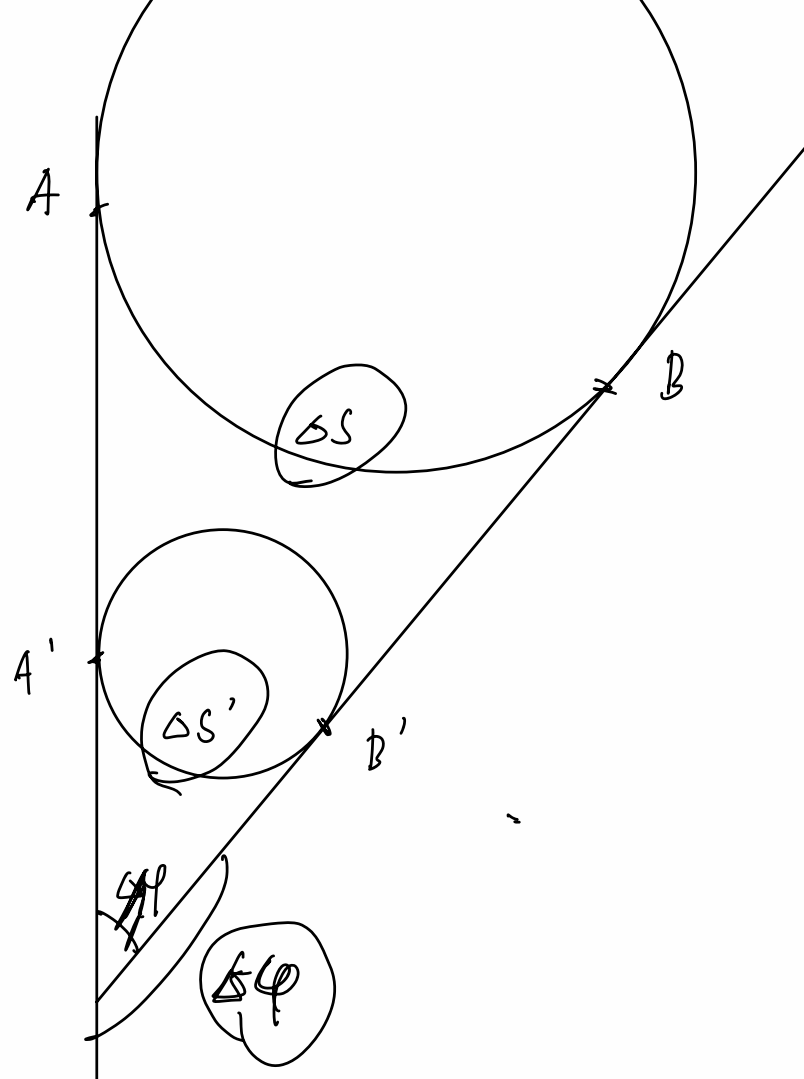
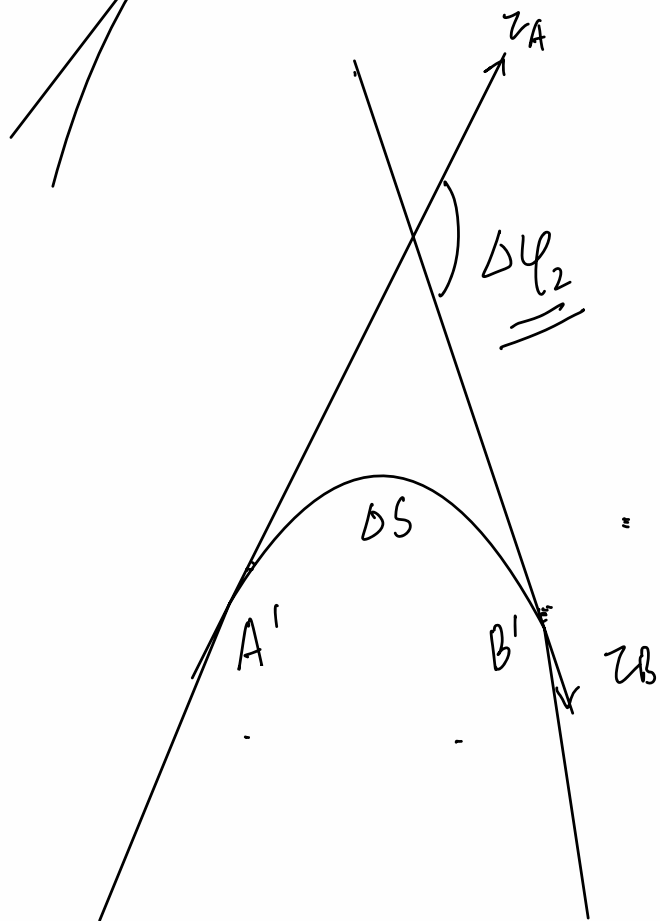
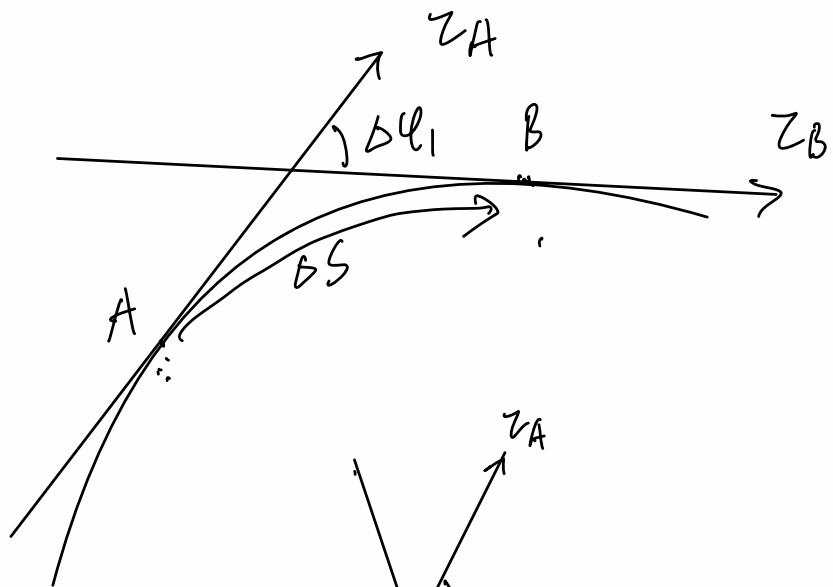
$[c, d]$

$$L = \int_c^d \sqrt{1 + (f^{-1})'(y)} \, dy.$$

例：计算曲线 $y = \left(\frac{x}{2}\right)^{\frac{2}{3}}$ 从 $x=0$ 到 $x=2$ 的弧长。



§ 平面曲线的曲率



$$\bar{k} = \frac{\Delta\varphi}{\Delta s} \quad \text{平均曲率.}$$

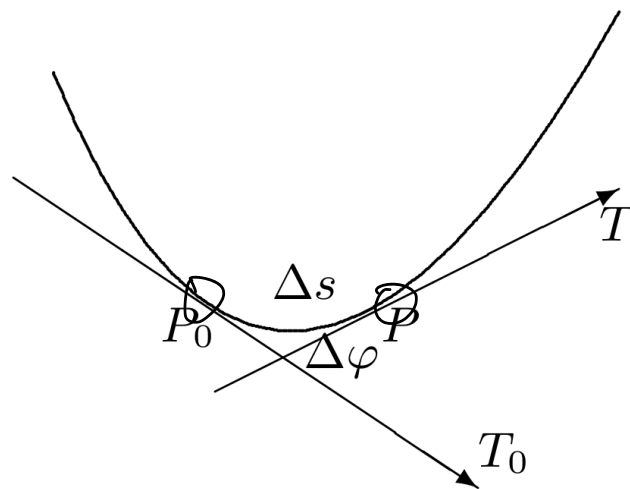
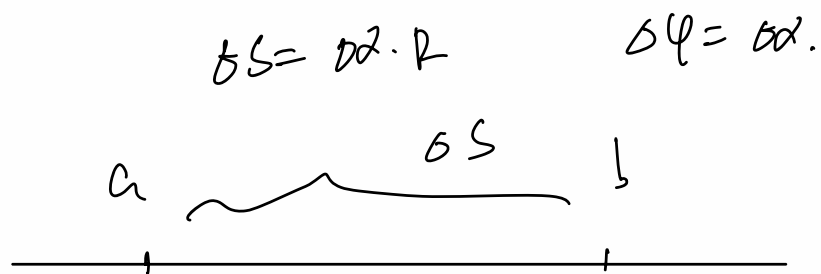


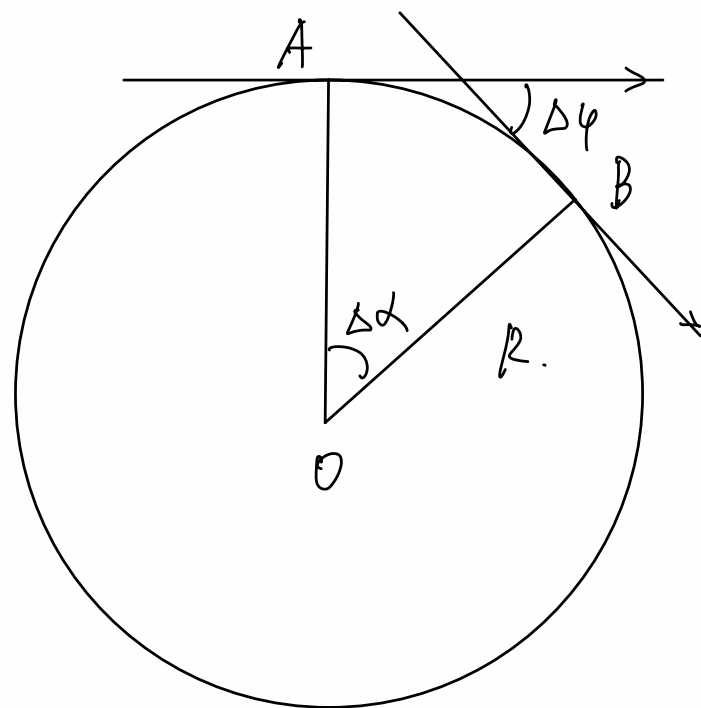
图 6.10 平面曲线曲率示意图

$$\bar{k} = \frac{\Delta\varphi}{\Delta s} = \frac{\Delta\alpha}{\Delta s} = \frac{1}{R}$$



$$\Delta\varphi = 0.$$

$$\bar{k} = \frac{\Delta\varphi}{\Delta s} = 0.$$



$$\lim_{\Delta s \rightarrow 0} \frac{\Delta\varphi}{\Delta s} = \lim_{\Delta s \rightarrow 0} \bar{k} = k \quad \text{在 } P_0 \text{ 点的曲率.}$$

$\delta s \rightarrow 0$ $\equiv$

$$K = \left| \frac{d\varphi}{ds} \right| = \left| \lim_{\delta s \rightarrow 0} \frac{\delta \varphi}{\delta s} \right|$$

$$[x, y]^t$$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}$$

$$\varphi(t)$$

$$s(t)$$

$$\frac{\varphi'(t)}{s'(t)}$$

$$\varphi(t) = \arctan \frac{y'(t)}{x'(t)}$$

$$d\varphi = \frac{x'(t)y''(t) - x''(t)y'(t)}{[x'(t)]^2 + [y'(t)]^2} dt$$

$$s(t) = \int_a^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2} d\tau$$

$$y = f(x)$$

$$ds = s'(t) dt = \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

$$K = \frac{|f''(x)|}{|1 + [f'(x)]^2|^{\frac{3}{2}}}$$

$$K = \frac{d\varphi}{ds} = \left| \frac{x'(t)y''(t) - x''(t)y'(t)}{[x'(t)]^2 + [y'(t)]^2} \right|$$

$$r = r(\theta)$$

$$K(\theta) = \frac{|r^2 + 2(r')^2 - r r''|}{|r^2 + (r')^2|^{\frac{3}{2}}}$$

例 6.2.4 求椭圆 $l: x = a \cos t, y = b \sin t, 0 \leq t \leq 2\pi$ ($a \geq b > 0$) 上各点处的曲率, 并指出曲率的最大值和最小值.

$$x'(t) = -a \sin t$$

$$x''(t) = -a \cos t$$

$$y'(t) = b \cos t$$

$$y''(t) = -b \sin t$$

$$k(t) = \left| \frac{y''x' - y'x''}{(x'^2 + y'^2)^{3/2}} \right| = \frac{ab}{|a^2 \sin^2 t + b^2 \cos^2 t|^{3/2}}$$

$$t=0$$

$$\frac{a}{b^2}$$

$$t=\frac{\pi}{2}$$

$$\frac{b}{a^2}$$

$$a=b$$

$$k = \frac{1}{a} = \frac{1}{R}$$

$$曲率 = \frac{1}{半径}$$

一般曲线 曲率半径 $R = \frac{1}{k}$

微元法 (Differential element method)

$y=f(x)$, 连续, Q 和 $f(x)$ 有关.

① Q 是在 $[a, b]$ 上的定积分, $Q([a, b])$.

考虑 $[a, x]$, ($x \in [a, b]$), Q 又依赖于 x 成为变量. $Q(x)$

②. $[a, b]$ 的任意子区间, $[\alpha, \beta] \subseteq [a, b]$, $[\beta, \gamma] \subseteq [a, b]$.

$$Q([\alpha, \gamma]) = Q([\alpha, \beta]) + Q([\beta, \gamma]).$$

③. $\Delta Q_i \sim f(\xi_i) \Delta x_i$

弧长微元

$$\begin{cases} \underline{\underline{ds}} = \sqrt{[x'(t)]^2 + [y'(t)]^2} dt \\ \underline{\underline{ds}} = \sqrt{1 + [f'(x)]^2} dx \end{cases}$$

$$\begin{aligned} S &= \int_a^x \dots \\ ds &= \end{aligned}$$

$f(x) \geq 0$ 在 $[a, b]$ 上连续.

任取 $x \in [a, b]$.

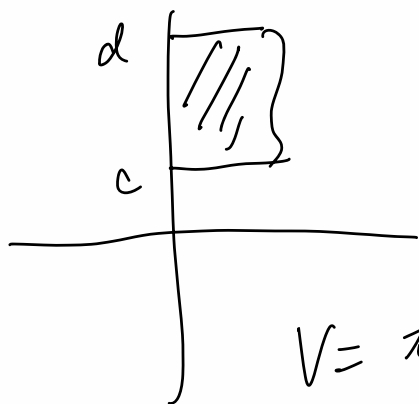
$[x, x+dx]$. $\{x=x\}$

$$dV = \pi f(x)^2 dx$$

$$V = \int_a^b \pi f(x)^2 dx$$

截面法.

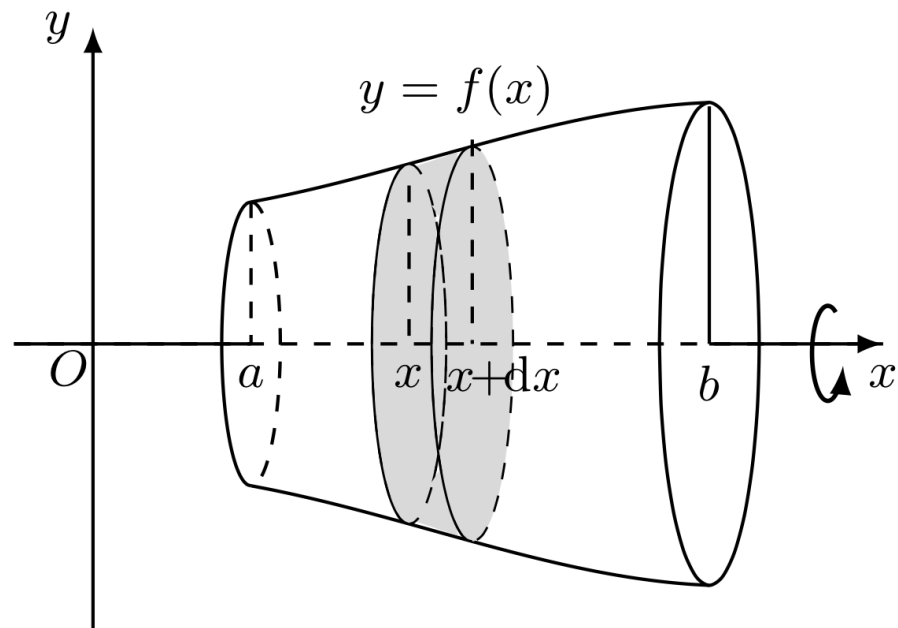
$x = \varphi(y)$, $y \in [c, d]$.



$$V = \pi \int_c^d (\varphi(y))^2 dy.$$

$f(x) = c \geq 0$.

$$\underline{\underline{r = f(x)}} \quad \pi f(x)^2$$



§ 截面积法

先分割.

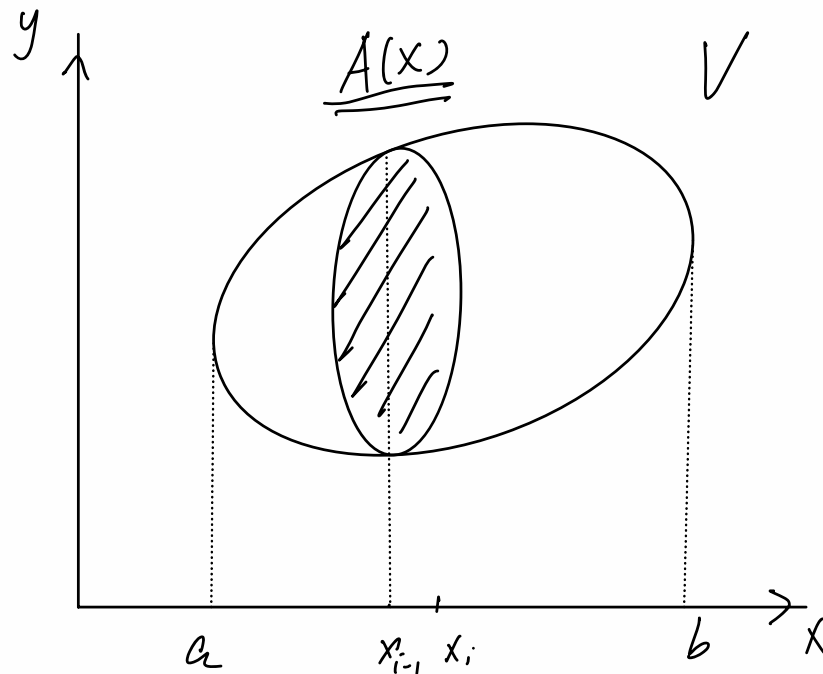
$$a = x_0 < x_1 < \dots < x_n = b,$$

$$\lambda = \max \Delta x_i$$

Δx_i

$$\xi_i \in [x_{i-1}, x_i].$$

$$\lim_{\lambda \rightarrow 0} \sum_{i=1}^n A(\xi_i) \Delta x_i = \int_a^b A(x) dx = V.$$



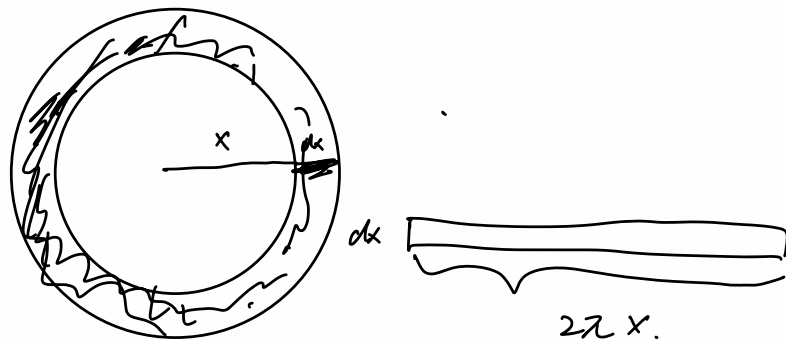
§ 柱壳法

$f(x) \geq 0$, 在 $[a, b] \subseteq \mathbb{R}$ 上连续.

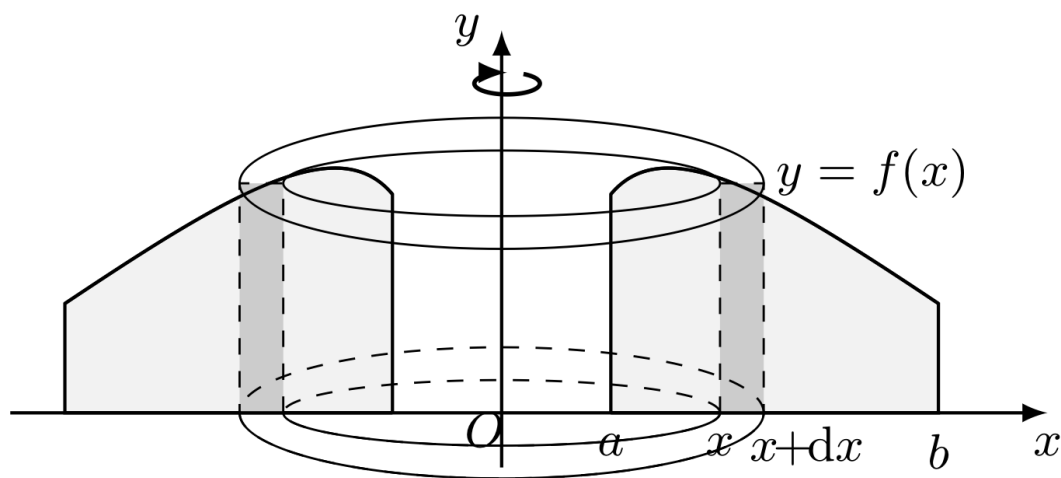
任取 $[x, x+dx]$.

$$\underline{\underline{2\pi x dx f(x) = dV}}$$

$$\underline{\underline{V = 2\pi \int_a^b x f(x) dx}}$$



$$\lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi [(x_i+dx)^2 - x_i^2] \cdot \underline{\underline{f(\xi_i)}}$$



例 6.3.1 设 D_1 是由抛物线 $y = 2x^2$ 和直线 $x = a$, $x = 2$ 及 $y = 0$ 所围成的平面区域, D_2 是由抛物线 $y = 2x^2$ 和直线 $y = 0$, $x = a$ 所围成的平面区域, 其中 $0 < a < 2$ (图 6.15).

(1) 求 D_1 绕 x 轴旋转而成的旋转体体积 V_1 ;

(2) 求 D_2 绕 y 轴旋转而成的旋转体体积 V_2 .

$$\begin{aligned} \text{①. } V &= \pi \int_a^2 y^2 dx = \pi \int_a^2 (2x^2)^2 dx \\ &= 4\pi \int_a^2 x^4 dx = \frac{4}{5} \pi \cdot x^5 \Big|_a^2 \\ &= \frac{4}{5} \pi (32 - a^5) \end{aligned}$$

$$\begin{aligned} \text{②. } V &= 2\pi \int_0^a x y dx \\ &= 2\pi \int_0^a x \cdot 2x^2 dx = 4\pi \int_0^a x^3 dx \\ &= \pi a^4 \quad \text{挖空法.} \end{aligned}$$

$$\underline{\underline{V'}} = \pi a^2 \cdot 2a^2 = 2\pi a^4$$

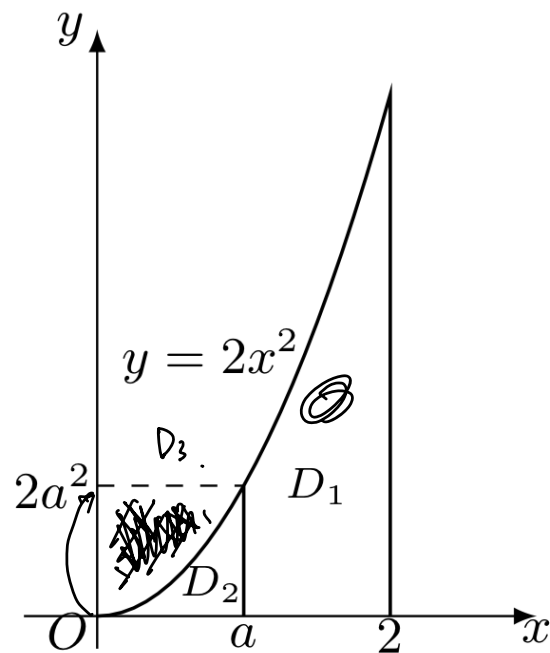
$$\underline{\underline{V''}} = \pi \int_0^{2a^2} \left(\sqrt{\frac{y}{2}} \right)^2 dy$$

$$= \frac{\pi}{2} \int_0^{2a^2} y^2 dy = \pi a^4$$

$$y = 2x^2$$

$$x = \sqrt{\frac{y}{2}}$$

$$V = V' - V'' = \pi a^4$$



例 6.3.2 计算由摆线 $x = a(t - \sin t)$, $y = a(1 - \cos t)$ 相应于 $0 \leq t \leq 2\pi$ 的一拱, 直线 $y = 0$ 所围成的图形分别绕 x 轴、 y 轴旋转而成的旋转体的体积.

$$V_x = \pi \int_0^{2\pi a} y^2 dx$$

$$x = x(t)$$

$$y = y(t)$$

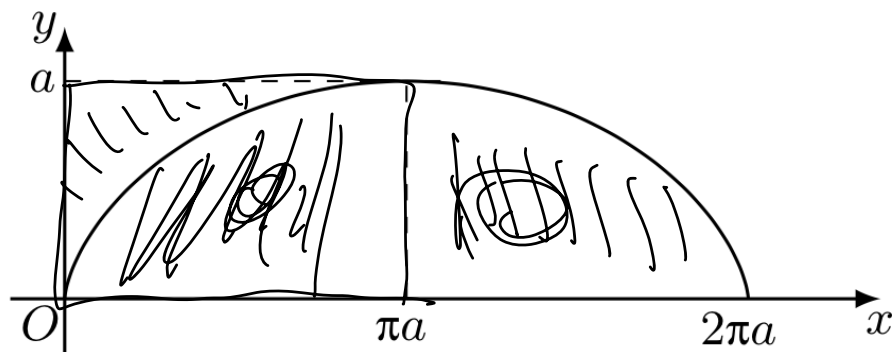
$$t \in [0, 2\pi]$$

$$= \pi \int_0^{2\pi} a^2 (1 - \cos t)^2 \cdot a (1 - \cos t) dt$$

$$= \pi a^3 \int_0^{2\pi} (1 - \cos t)^3 dt$$

$$= \pi a^3 \int_0^{2\pi} (1 - 3\cos t + 3\cos^2 t - \cos^3 t) dt$$

$$= 5\pi^2 a^3$$



$$V_y = 2\pi \int_0^{2\pi a} x y dx = 2\pi \int_0^{2\pi} a(t - \sin t) \cdot a(1 - \cos t) \cdot a(1 - \cos t) dt$$

$$= 2\pi a^3 \int_0^{2\pi} (t - \sin t) (1 - \cos t)^2 dt$$

$$\int_a^b f(x) dx = \int_{\frac{a+b}{2}}^{\frac{a+b}{2}} (f(x) + f(\frac{a+b}{2} - x)) dx$$

$$= 2\pi a^3 \int_0^{\pi} [(t - \sin t) (1 - \cos t)^2 + (2\pi - t + \sin t) (1 - \cos t)^2] dt$$

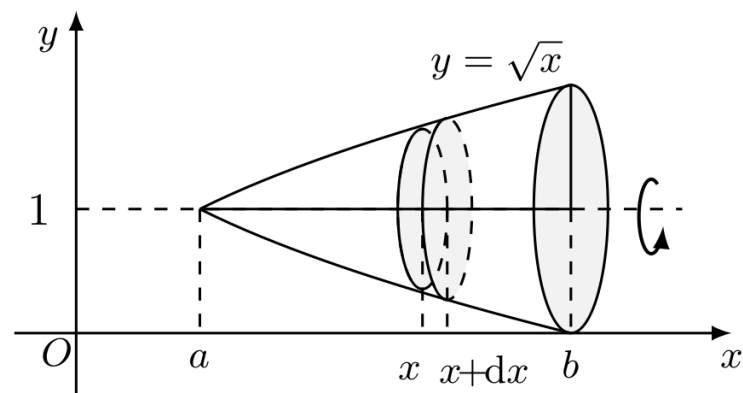
$$\sin(2\pi - t)$$

$$= -\sin t$$

$$\cos(2\pi - t) = \cos t$$

$$\begin{aligned}
 &= \frac{a^3 \int_0^{2\pi} (1 - \cos t)^2 dt}{\int_0^{2\pi} (1 - 2\cos t + \cos^2 t) dt} \\
 &= 4\pi^2 a^3 \cdot \frac{3}{2} \pi = 6\pi^3 a^3
 \end{aligned}$$

例 6.3.3 求由 $y = \sqrt{x}$ 和直线 $y = 1, x = 4$ 所围成的区域绕直线 $y = 1$ 旋转一周所生成立体的体积.



求旋转体的面积

$f(x) \geq 0$ on $[a, b]$. $f(x)$ 可导, $f'(x)$ 连续.

任取 $[x, x+dx]$.

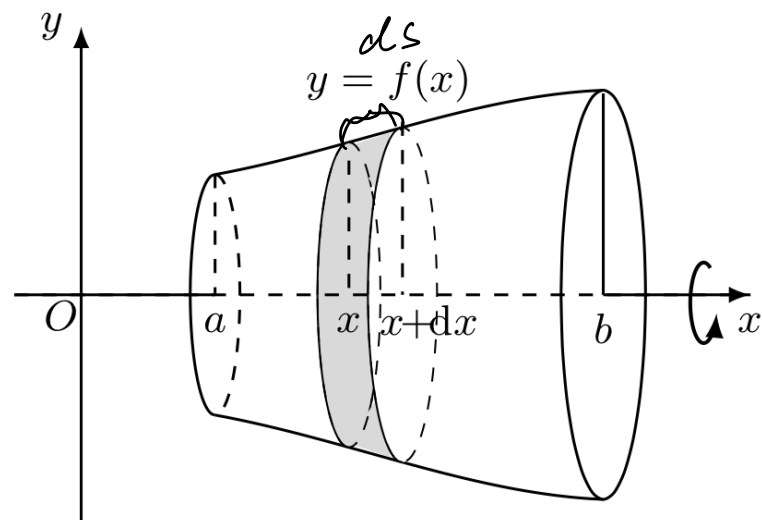
$$r = f(x), \quad C = 2\pi f(x).$$

$$dS = \sqrt{1 + (f'(x))^2} dx$$

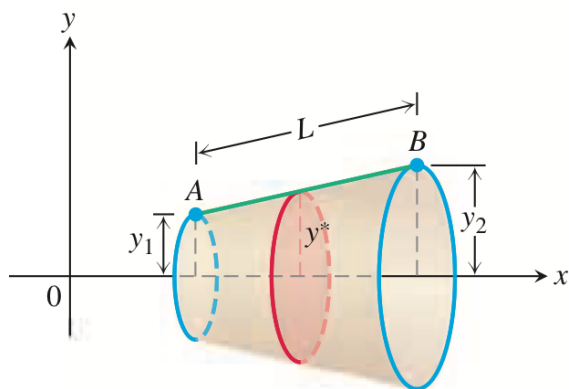
$$S = C \cdot h$$

$$= 2\pi f(x) \cdot \sqrt{1 + (f'(x))^2} dx = dA$$

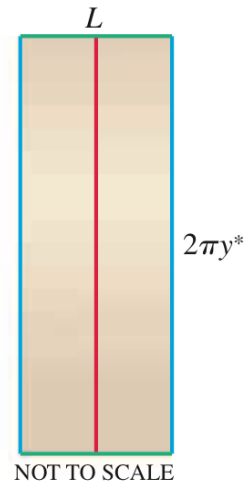
$$A = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx.$$



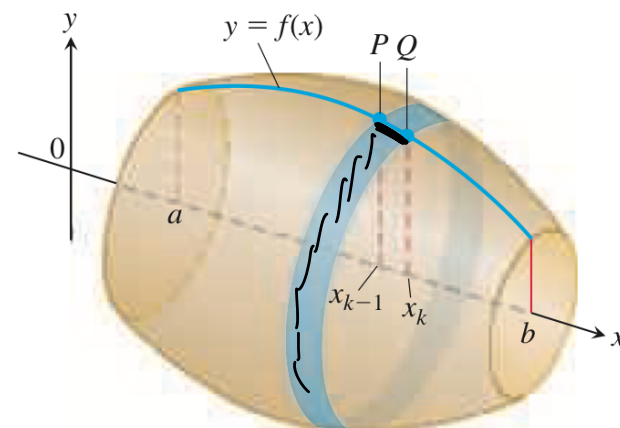
$$C = x_0 < x_1 < \dots < x_n = b.$$



(a)



(b)



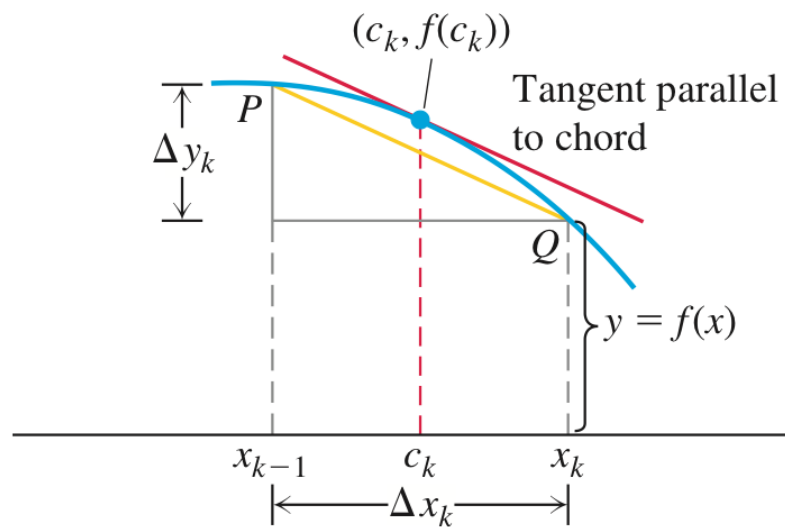
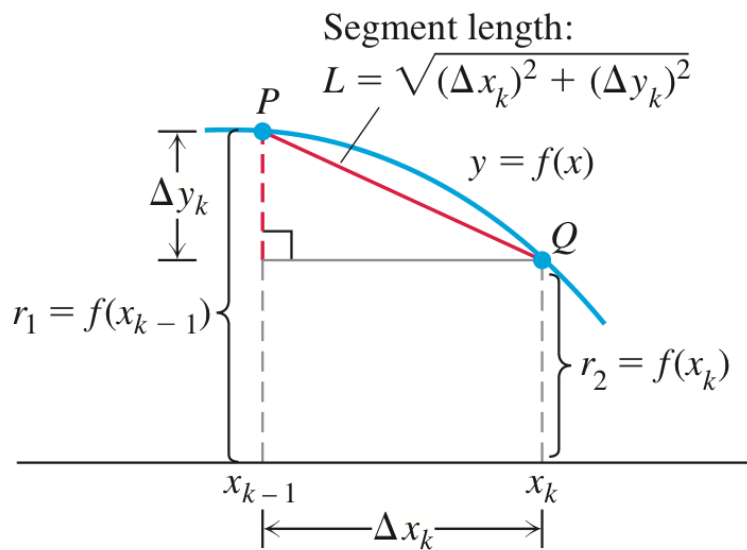
$$f(x) \sqrt{1 + (f'(x))^2}$$

裁断体

$$\int \pi (f(x_{k-1}) + f(x_k)) \cdot ds$$

组合中值.

26 1.1.1



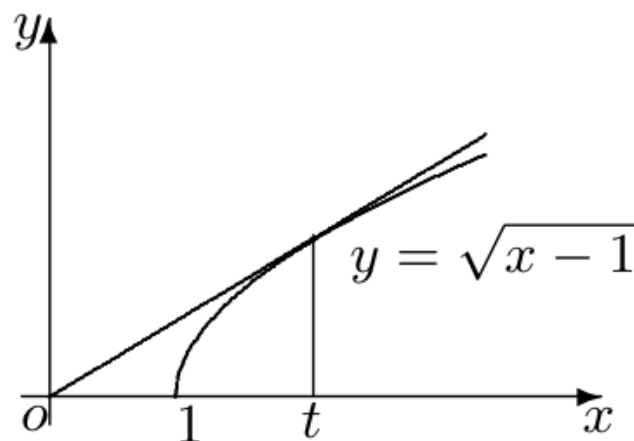
例 6.3.4 设有曲线 $y = \sqrt{x-1}$, 过原点作其切线, 求此曲线、切线及 x 轴围成的平面图形绕 x 轴旋转一周所得到的旋转体的表面积.

$$\underline{(t, \sqrt{t-1})} \quad k = \frac{dy}{dx} = \frac{1}{2\sqrt{x-1}}$$

$$y = \frac{1}{2\sqrt{t-1}} x$$

$$\sqrt{t-1} = \frac{1}{2\sqrt{t-1}} t.$$

$$t=2.$$



$$(2, 1) \quad y = \frac{x}{2} \quad [0, 2].$$

$$\underline{y = \sqrt{x-1}} \quad [1, 2].$$

$$A_1 = 2\pi \int_1^2 \sqrt{x-1} \cdot \sqrt{1 + \left(\frac{1}{2\sqrt{x-1}}\right)^2} dx$$

$$= \pi \int_1^2 \sqrt{4x-3} dx = \frac{\pi}{6} (5\sqrt{5} - 1)$$

$$\frac{dy}{dx} = \frac{1}{2}$$

$$A_2 = \pi \int_0^2 \frac{1}{2} x \sqrt{1 + \left(\frac{1}{2}\right)^2} dx$$

$$x = \varphi(y) \geq 0. \quad [c, a].$$

$$= \frac{\sqrt{5}}{2} \pi \int_0^2 x \, dx = \sqrt{5} \pi$$

$$A_1 + A_2 = \frac{\pi}{6} (11\sqrt{5} - 1)$$

$$S = \int_c^d 2\pi \cdot x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

例：求直线 $x = 1 - y$, $0 \leq y \leq 1$, 绕 y 轴旋转的表面积。

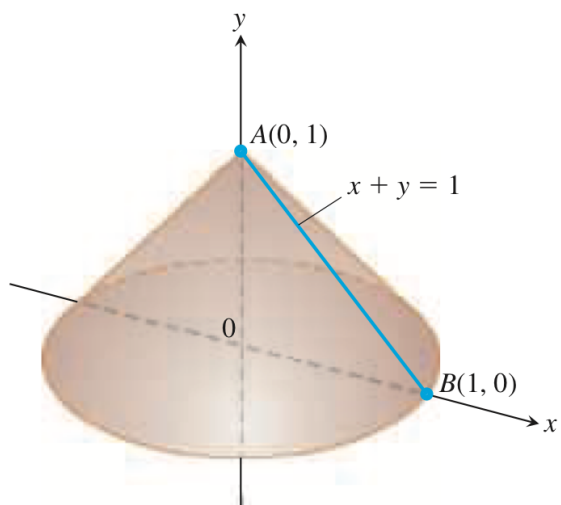
$$A = \pi \cdot r \cdot l$$

$$= \pi \cdot 1 \cdot \sqrt{2} = \sqrt{2} \pi$$

$$\frac{dx}{dy} = -1. \quad \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{2}.$$

$$A = \int_0^1 2\pi \cdot x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= \int_0^1 2\pi (1-y) \sqrt{2} dy$$



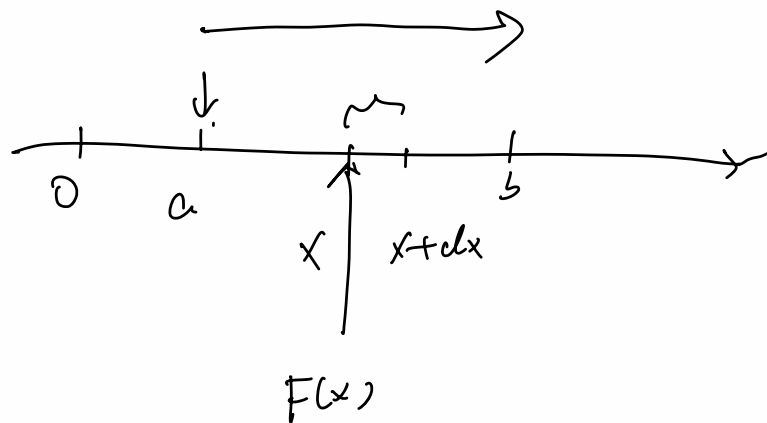
$$= \lambda \sqrt{L} \int_0^L (1-y) dy = \sqrt{L} \lambda.$$

m.

变力做功. $F = C \ddot{q} b.$

$$W = F(b-a)$$

$$F = F(x) \quad (a \leq x \leq b)$$



$$dW = F(x) dx$$

$$W = \int_a^b F(x) dx.$$

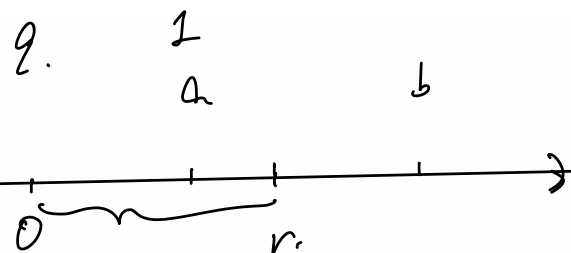
变为做功

例 6.4.1 将一个电量为 q 的正电荷放在 x 轴的坐标原点处，它产生的电场将对周围的电场有作用力，试求将一个单位电荷从 $x = a$ 处沿 x 轴移动到 $x = b$ 处时，电场所作的功，其中 $ab > 0$.

$$F = k_e \cdot \frac{q_1 q_2}{r^2}$$

$$8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2.$$

$$\frac{1}{4\pi\epsilon_0} \leftarrow \frac{1}{4\pi\epsilon_0} \text{ 真空介电常数}$$



$$F(x) = k \cdot \frac{q}{x^2}$$

$$W = \int_a^b F(x) dx$$

$$= kq \int_a^b \frac{1}{x^2} dx$$

$$= kq \left(\frac{1}{a} - \frac{1}{b} \right)$$

~~~~~

$$W = \int_a^{+\infty} F(x) dx = \frac{kq}{a}.$$

**例 6.4.2** 为清除井底的污泥，用缆绳将抓斗放入井底抓起污泥后提出井口．已知井深 30 m，抓斗自重 400 N，缆绳每米 50 N，抓斗抓起污泥重 2000 N，提升速度为 3 m/s，在提升过程中，污泥以 20 N/s 的速率从缝隙中漏掉，现将抓起污泥的抓斗提升至井口，问克服重力需作多少焦耳的功？(说明：不计抓斗的高度及位于井口上方的缆绳； $1 \text{ N} \times 1 \text{ m} = 1 \text{ J}$ , m, N, s, J 分别表示米、牛顿、秒、焦耳．)

设取  $h$  时 -  $p = \rho g h$   $P = p \cdot A$

液体的压力。

例 6.4.3 某闸门的形状与大小如图 6.20 所示, 其中直线  $l$  为对称轴, 闸门的上部为矩形  $ABCD$ , 下部由抛物线与线段  $AB$  所围成, 当水面与闸门上端相平时, 欲使闸门矩形部分承受的水压力与闸门下部承受的水压力之比为 5:4, 闸门矩形部分的高  $h$  应为多少  $m$ (米)?

$$[y, y+dy] \quad [1, h+1].$$

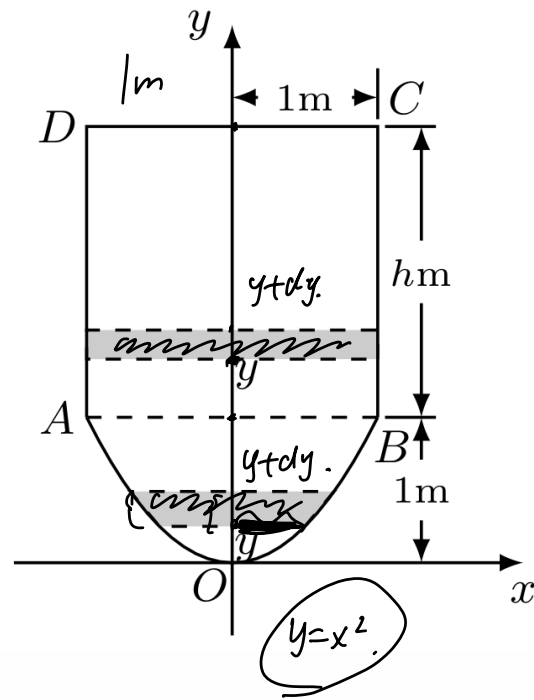
$$dP = \rho g (h+1-y) \cdot \underline{\underline{2 \cdot dy}}$$

$$P = \int_1^{h+1} \rho g (h+1-y) \cdot 2 dy. \quad \underline{\underline{= \rho g h^2.}}$$

$$dP = \rho g (h+1-y) \cdot 2\sqrt{y} dy$$

$$P = \int_0^1 \rho g (h+1-y) 2\sqrt{y} dy = \underline{\underline{4\rho g \left( \frac{1}{3}h + \frac{2}{15} \right)}} = \frac{4}{5} \rho g h^2.$$

$$\begin{cases} h=2 \\ h=-\frac{1}{3} \end{cases}$$



例：从地面垂直发射质量为  $m$  的物体，计算物体从 A 点飞到 B 点  
 地球引力的做功。若要让物体飞离地球引力范围，  
 物体的初速度应设多少？

第二宇宙速度

$$m_1, m_2, r, F = k \frac{m_1 m_2}{r^2}$$

$$F = k \frac{Mm}{r^2}$$

$M$  地球质量

$r$  球心到物体的距离

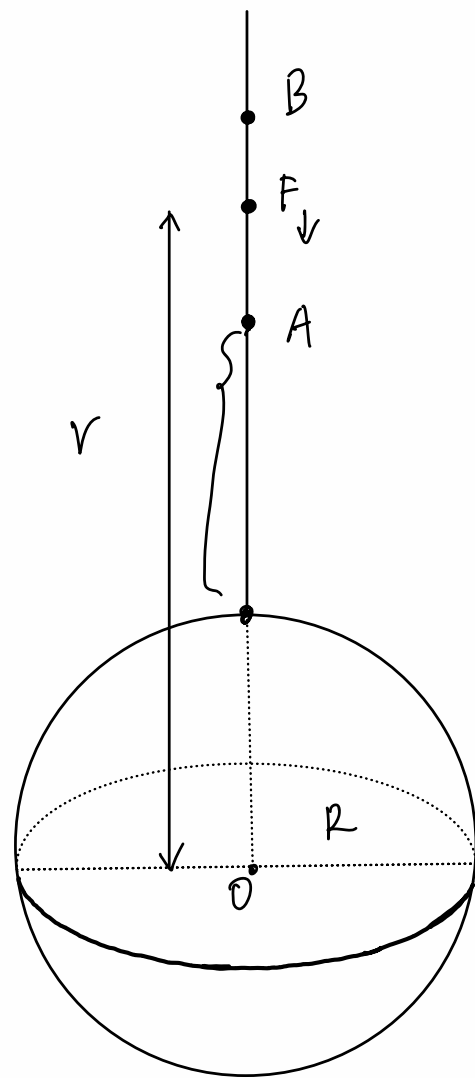
$$F = k \frac{Mm}{R^2} = mg$$

$$k = \frac{R^2 g}{M}$$

$$F = \frac{R^2 g}{R^2} \cdot \frac{Mm}{r^2} = mg \frac{R^2}{r^2}$$

$$W = \int_{R_1}^{R_2} (-F) dr = - \int_{R_1}^{R_2} mg \frac{R^2}{r^2} dr$$

$R_2 \rightarrow \infty$



$$= mgR^2 \left. \frac{1}{r} \right|_{R_1}^{(R_2)} = \underline{mgR^2 \left( \frac{1}{R_2} - \frac{1}{R_1} \right)}$$

$$W' = \lim_{R_2 \rightarrow \infty} W = -\frac{1}{R} mgR^2 = -mgR.$$

$$\underline{mgR = \frac{1}{2} m v^2}$$

$$v = \sqrt{2gR}.$$

$$g = 9.8.$$

$$R = 6371 \text{ km.}$$

$$= 6.37 \times 10^6 \text{ m}$$

$$\underline{v = 11.2 \text{ km/s}}$$